

A Recursive Notation over $\{1, 2, 3\}$ for Integers Coprime to Six, with an Application to Difference-of-Squares Factorisation

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Abstract

We define a canonical notation writing every positive integer using only the atoms $1, 2, 3$, multiplication, and the operations ± 1 . The construction rests on the observation that every integer coprime to 6 has the form $2 \cdot 3 \cdot k \pm 1$ for a unique k and a forced sign. Applying the rule recursively to k terminates, since each level divides the argument by approximately six. We prove that the representation is unique once prime factorisation is applied first, and that it is self-verifying: the expression evaluates to the number it denotes. We then record a companion identity for differences of squares and the enumeration of Pythagorean partners it yields.

1 The $6k \pm 1$ notation

Definition 1 (Atoms and forms). An *expression* is built from the atoms $1, 2, 3$ by multiplication and by the two unary operations $x \mapsto 2 \cdot 3 \cdot x - 1$ and $x \mapsto 2 \cdot 3 \cdot x + 1$.

Proposition 1 (Trichotomy modulo six). Let n be a positive integer with $\gcd(n, 6) = 1$. Then exactly one of

$$n = 6k - 1, \quad n = 6k + 1$$

holds, with k a uniquely determined positive integer.

Proof. The residues coprime to 6 are 1 and 5. If $n \equiv 1$, then $k = (n - 1)/6$; if $n \equiv 5$, then $k = (n + 1)/6$. In each case k is an integer and the other equation has no integral solution. $\square$

Proposition 2 (Closure and sign parity). The set $\{n : \gcd(n, 6) = 1\}$ is closed under multiplication, and

$$(6a - 1)(6b - 1) \equiv +1, \quad (6a - 1)(6b + 1) \equiv -1, \quad (6a + 1)(6b + 1) \equiv +1 \pmod{6}.$$

Consequently $(6k - 1)^r \equiv (-1)^r$ and $(6k + 1)^r \equiv +1 \pmod{6}$ for all $r > 0$.

Theorem 1 (Existence, uniqueness, termination). Every positive integer n admits exactly one expression obtained by the rule

1. factor n into primes and write the expression as the product of the expressions of its prime factors;
2. for a prime $p \leq 3$, the expression is the atom itself;
3. for a prime $p > 3$, write $p = 6k \pm 1$ with the sign forced by $p \bmod 6$, and recurse on k .

The recursion terminates.

Proof. Uniqueness of step 1 is the fundamental theorem of arithmetic; uniqueness of step 3 is the previous proposition. For termination, note $k < p/5$ for $p > 3$, so the argument strictly decreases at each level and the descent reaches an atom in $O(\log n)$ steps. $\square$

Example 1.

$$\begin{aligned} 41 &= (2 \cdot 3 \cdot (2 \cdot 3 + 1) - 1), & 25 &= (2 \cdot 3 - 1) \cdot (2 \cdot 3 - 1), \\ 35 &= (2 \cdot 3 - 1) \cdot (2 \cdot 3 + 1), & 97 &= (2 \cdot 3 \cdot (2 \cdot 2 \cdot 2 \cdot 2) + 1). \end{aligned}$$

Proposition 3 (Structural reading). In the canonical expression of n , the outermost operation is a product if and only if n is composite, and is ± 1 if and only if n is prime.

remark 1. The notation is a *certificate*, not a test. Verifying a given expression costs $O(\log n)$ arithmetic operations, whereas producing it requires the prime factorisation of n . The ± 1 layer carries no multiplicative information: for any $n = 6k \pm 1$ one has $n \equiv \pm 1 \pmod{6}$, so no level of the descent shares a factor with the level below it.

Proposition 4 (Perfect-power splittings). If $n + 1 = s^m$ then $(s - 1) \mid n$, and if $n - 1 = s^m$ with m odd then $(s + 1) \mid n$, by

$$s^m - 1 = (s - 1) \sum_{j=0}^{m-1} s^j, \quad s^m + 1 = (s + 1) \sum_{j=0}^{m-1} (-1)^j s^j.$$

Example 2. $215 = 6^3 - 1 = 5 \cdot 43$ and $217 = 6^3 + 1 = 7 \cdot 31$.

2 The difference-of-squares identity

Proposition 5 (Gap identity). For $B = A + d$,

$$B^2 - A^2 = 2Ad + d^2 = d(2A + d).$$

corollary 1. For odd n , every factorisation $n = d(2A + d)$ corresponds to a divisor $d \mid n$ with $A = (n - d^2)/(2d)$, and conversely. In particular $d = 1$ always yields the trivial factorisation $n = 1 \cdot n$.

Proposition 6 (Two parametrisations). The equation $n = ab$ may be solved by fixing either unknown:

fix d and solve for $A \implies$ trial division, fix x and solve for y in $n = x^2 - y^2 \implies$ Fermat's method.

The first is efficient when n has a small factor, the second when the factors are close together; the cost of Fermat's method is $\approx y^2/(2\sqrt{n})$ steps.

Proposition 7 (Wheel restriction). If $\gcd(n, 6) = 1$ then every divisor of n is of the form $6k \pm 1$. Hence a trial-division search need only test candidates in these two residue classes, eliminating four of every six.

3 Pythagorean partners

Definition 2. For a positive integer A , a *partner* of A is a pair (B, C) of positive integers with $B^2 - A^2 = C^2$.

Theorem 2 (Enumeration of partners). The partners of A are in bijection with the factorisations $A^2 = de$ with $d < e$ and $d \equiv e \pmod{2}$, via

$$B = \frac{d+e}{2}, \quad C = \frac{e-d}{2}.$$

Proof. $B^2 - C^2 = (B - C)(B + C) = de = A^2$. Conversely, given $B > C > 0$ with $B^2 - C^2 = A^2$, set $d = B - C$ and $e = B + C$; then $de = A^2$ and $d \equiv e \pmod{2}$ since B, C are integers. $\square$

corollary 2. The number of partners of A is determined by the divisor count of A^2 . In particular A prime admits exactly one partner, namely $((A^2 + 1)/2, (A^2 - 1)/2)$.

Proposition 8 (Sum–difference conversion). The identity

$$p^2 + q^2 = n^2 \iff (p - q)^2 + (p + q)^2 = 2n^2$$

converts Pythagorean triples with hypotenuse n into representations of $2n^2$ as a sum of two squares.

4 Implementation

```
from math import isqrt

def smallest_factor(n):
    for d in (2, 3):
        if n % d == 0:
            return d
    k = 1
    while True:
        for d in (6*k - 1, 6*k + 1):
            if d * d > n:
                return n
            if n % d == 0:
                return d
        k += 1

def rep(n):
    if n <= 3:
        return str(n)
    p = smallest_factor(n)
    if p != n:
        return rep(p) + "*" + rep(n // p)
    k = (n + 1) // 6 if (n + 1) % 6 == 0 else (n - 1) // 6
    s = "-" if (n + 1) % 6 == 0 else "+"
    core = "2*3" if k == 1 else "2*3*" + rep(k)
    return "(" + core + s + "1)"

def partners(A):
    out, n = [], A * A
```

```

for d in range(1, isqrt(n) + 1):
    if n % d == 0:
        e = n // d
        if d < e and (d + e) % 2 == 0:
            out.append(((d + e) // 2, (e - d) // 2))
return out

```

Proposition 9 (Verification). For all n with $\gcd(n, 6) = 1$ and $n < 10^5$, the expression produced by `rep` evaluates to n , and no two distinct n produce the same expression. Expression lengths in this range have maximum 55 characters (attained at $n = 96875 = 5^5 \cdot 31$) and mean 40.3.

5 Positional interpretation: base six with a signed unit digit

Proposition 10 (Digit reading). Let $\gcd(n, 6) = 1$ and let $n = 6k + \varepsilon$ with $\varepsilon \in \{-1, +1\}$ forced by $n \bmod 6$. Then the descent

$$n \mapsto k \mapsto \dots$$

is a positional expansion in base six in which the unit digit is written ± 1 rather than as a residue in $\{1, 5\}$, and in which the leading part is re-expanded by the same rule.

Example 3. $41 = 105_6$, and $105_6 = 6 \cdot 7 - 1$ with $7 = 11_6 = 6 \cdot 1 + 1$, giving $(2 \cdot 3 \cdot (2 \cdot 3 + 1) - 1)$.

corollary 3 (Depth bound). The descent applied to n terminates after at most $\lceil \log_6 n \rceil$ levels.

6 Minimality of the canonical form

Definition 3. The *size* of an expression is its number of atoms plus its number of operations.

Theorem 3 (Shortest descent). Among all expressions for a prime $p > 3$ built from the atoms $\{1, 2, 3\}$ by multiplication and ± 1 , the canonical form obtained by the rule $p = 6k \pm 1$ has minimal depth.

Proof. Any expression for p ending in ± 1 has the form $mq \pm 1$ with m a product of atoms; since $p \equiv \pm 1 \pmod{6}$ and $2, 3 \nmid p$, the largest admissible m is 6. Every level therefore reduces the argument by a factor of at most six, and the canonical rule attains this bound at every level. Induction on n gives the claim. $\square$

remark 2. The count of expressions of a given size may be obtained by dynamic programming over the grammar $E \rightarrow 2 \mid 3 \mid E \times E \mid E + 1 \mid E - 1$, tabulating for each size the multiset of representable values. Without a size bound the count is infinite, since $n \mapsto (n + 1) - 1$ pads any expression.

7 The coprimality lemma

Lemma 1 (Levels are coprime). Let $n = 6k \pm 1$. Then

$$n \equiv \pm 1 \pmod{k}, \quad \text{hence} \quad \gcd(n, k) = 1.$$

Proof. Immediate from $n \mp 1 = 6k$. $\square$

corollary 4 (Negative results). Lemma 1 excludes the following as factorisation aids:

1. *Inner divisors.* No value appearing at a level of the descent below n divides n by construction; the level immediately below leaves remainder ± 1 exactly.
2. *Neighbour tests.* Since $n = 6(k \mp 1) \pm 7$ and $n = 6(k \pm 1) \mp 5$, one has $n \bmod (k \pm 1) \in \{5, 7, -5, -7\}$, so $k \pm 1$ divides n only when $k \pm 1 \in \{5, 7\}$, i.e. only for $n \in \{25, 35, 49\}$.
3. *Sign counting.* The descent of n has at most $\lceil \log_6 n \rceil$ levels, so any statistic computed from the multiset of signs takes $O(\log^2 n)$ distinct values, whereas $\pi(n) \sim n/\log n$. By the pigeonhole principle no such statistic separates primes from composites.

Example 4. 11 and 35 both have canonical forms with one -1 and no $+1$; 37 and 25 both have one $+1$ and no -1 . In each pair one entry is prime and the other is not.

8 Expression statistics

Proposition 11 (Measured lengths). Over all n with $\gcd(n, 6) = 1$ and $n < 10^5$ (a set of 33,333 integers), the canonical expression produced by `rep` satisfies:

maximum length	55 characters, attained at $n = 96875 = 5^5 \cdot 31$
mean length	40.3 characters
evaluation	$\text{eval}(\text{rep}(n)) = n$ for every such n
collisions	none: <code>rep</code> is injective on this range

remark 3. Length is driven by two independent quantities: the number of prime factors, whose expressions are concatenated, and the depth of each factor's descent, which is $O(\log p)$. The observed growth is consistent with $O(\Omega(n) \log n)$, where Ω counts prime factors with multiplicity. The extremal case $5^5 \cdot 31$ is extremal for the first reason rather than the second.

9 Limits of the notation

Proposition 12 (Primality is visible but not computable from the form). Let $n > 3$ with $\gcd(n, 6) = 1$. In the canonical expression of n , the outermost operation is a product if and only if n is composite. However, producing that expression requires the prime factorisation of n , and verifying a given expression costs $O(\log n)$ operations.

remark 4 (Certificate, not test). The notation is therefore a *certificate format*. It records the complete multiplicative structure of n in a string over $\{1, 2, 3\}$ that is self-verifying: no external table, no oracle, and no trust in the producer are required to check it. This is a strictly weaker property than decidability of the structure, and the distinction is not repaired by any re-encoding, since by Corollary 4 the additive ± 1 layer and the multiplicative layer share no information.

remark 5 (Comparison with primality testing). Primality itself lies in P, and in practice Miller–Rabin decides it in $O(\log^3 n)$ time without producing any factorisation. A practical pipeline therefore establishes primality first and generates the canonical expression second, rather than inferring primality from the expression.

10 Machine learning on the notation

10.1 Decomposition of the task

Generating $\text{rep}(n)$ from the decimal digits of n splits into two subtasks of markedly different difficulty.

The descent. Given that p is prime, emitting $(2 \cdot 3 \cdot \text{rep}(k) \pm 1)$ requires only division by six and a residue read. This is a local, positional operation on digit strings, of depth $O(\log_6 p)$.

The split. Deciding whether the outermost operation is a product or a ± 1 requires the factorisation of n . Nothing in the digit representation of n determines it.

remark 6 (Why the second subtask resists learning). Three obstacles compound. First, coverage: a training set over $n < 10^5$ contains fewer than 10^4 distinct primes, whereas the prime factors of a 25-digit integer are drawn from more than 10^{21} candidates, so almost every large factor is a token unseen in training. Second, length generalisation: transformer models are not reliably able to extend multiplication beyond their training digit width, and factorisation is not easier than multiplication. Third, the labelling bottleneck: training data can only be generated as far as a classical factoring algorithm can label it.

10.2 The verification asymmetry

Proposition 13. Let s be a candidate expression over $\{1, 2, 3, \times, +1, -1\}$. Deciding whether s denotes n costs $O(|s|)$ arithmetic operations, independent of whether s is correct and independent of the difficulty of producing s .

This asymmetry is the property that makes the notation useful as a machine learning target: an incorrect proposal is rejected at negligible cost, so a model may be trained against a hard correctness signal rather than a token-level surrogate, and may be sampled repeatedly without risk of accepting a wrong answer.

```
def accept(s, n):  
    """Reject a proposed factored form in  $O(|s|)$ ."""  
    try:  
        return parse(s) == n and top_level_product(s)  
    except Exception:  
        return False
```

10.3 Proposed experiments

We state three experiments, each measurable and each avoiding the claim that a model can learn to factor.

1. *Descent only.* Train on pairs $(p, \text{rep}(p))$ for primes p , with primality supplied. Measure exact-match accuracy as a function of digit width beyond the training range. This isolates the learnable subtask and establishes an upper bound on what the architecture can contribute.
2. *Divisor ordering.* Use a model to rank trial divisors for composite n , and measure the expected number of probes to a factor against the ascending $6k \pm 1$ wheel on inputs whose factorisation is already known. A ranking that reduces the probe count is evidence of learned signal; the baseline is exact and the comparison is not confounded.

3. *Parameter prediction.* Predict a starting offset for Fermat's x , or the polynomial constant c in Pollard's rho, and measure the change in expected step count. Both are heuristics whose correctness does not depend on the prediction.

remark 7. None of these experiments requires a model to solve integer factorisation, and a negative result in each is informative. Claims to the contrary should be regarded with the scepticism appropriate to a problem on whose hardness widely deployed cryptography depends.

11 The two branches and their union

11.1 Definition of the branches

Definition 4 (Branches). Define

$$\mathcal{B}^- = \{6k - 1 : k \geq 1\}, \quad \mathcal{B}^+ = \{6k + 1 : k \geq 1\},$$

and write $\mathcal{B} = \mathcal{B}^- \cup \mathcal{B}^+$. We refer to $\mathcal{B}^-$ and $\mathcal{B}^+$ as the *lower* and *upper* branch.

11.2 The union is exactly the integers coprime to six

Theorem 4 (Branch characterisation). For an integer $n > 1$,

$$n \in \mathcal{B} \iff 2 \nmid n \text{ and } 3 \nmid n \iff \gcd(n, 6) = 1.$$

Moreover $\mathcal{B}^- \cap \mathcal{B}^+ = \emptyset$, so the union is disjoint and every element lies on exactly one branch.

Proof. Every integer is congruent to exactly one of $0, 1, 2, 3, 4, 5$ modulo 6. The residues $0, 2, 4$ are precisely those with $2 \mid n$; the residues $0, 3$ are precisely those with $3 \mid n$. Removing both families leaves the residues 1 and 5, which are $\mathcal{B}^+$ and $\mathcal{B}^-$ respectively. Disjointness holds because a single integer has a single residue. Finally, $\gcd(n, 6) = 1$ is equivalent to $2 \nmid n$ and $3 \nmid n$ since $6 = 2 \cdot 3$. $\square$

corollary 5 (Density). The branches have natural density $\frac{1}{6}$ each, so $\mathcal{B}$ has density $\frac{1}{3}$: a trial-division search restricted to $\mathcal{B}$ tests two candidates in every block of six, eliminating four of every six.

11.3 Contents of the union

Proposition 14 (Primes lie on the branches). Every prime $p > 3$ belongs to $\mathcal{B}$.

Proof. If $p \notin \mathcal{B}$ then by Theorem 4 either $2 \mid p$ or $3 \mid p$, contradicting primality for $p > 3$. $\square$

Proposition 15 (Closure under multiplication). $\mathcal{B} \cup \{1\}$ is closed under multiplication, with the sign rule

$$\varepsilon(mn) = \varepsilon(m)\varepsilon(n), \quad \varepsilon(6k \pm 1) := \pm 1.$$

Explicitly,

$$(6a-1)(6b-1) = 6(6ab-a-b)+1, \quad (6a-1)(6b+1) = 6(6ab+a-b)-1, \quad (6a+1)(6b+1) = 6(6ab+a+b)+1.$$

corollary 6 (Branch group). The map $\varepsilon : \mathcal{B} \rightarrow \{\pm 1\}$ is a surjective group homomorphism from $(\mathbb{Z}/6\mathbb{Z})^\times$ onto $\mathbb{Z}/2\mathbb{Z}$; the branches are its two cosets. Consequently a product stays on the lower branch if and only if an odd number of its factors lie there.

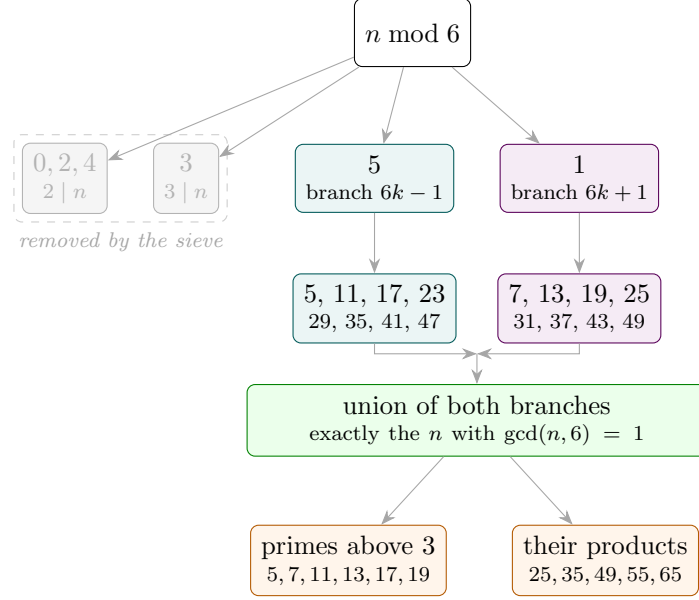


Figure 1: The residues modulo six. Four classes are eliminated by divisibility by 2 or 3; the two survivors are the branches $\mathcal{B}^-$ and $\mathcal{B}^+$, whose disjoint union is exactly $\{n : \gcd(n, 6) = 1\}$ — the primes above 3 together with all their products.

Proposition 16 (The union is not the set of primes). $\mathcal{B}$ contains every prime above 3 together with every product of such primes, and nothing else. The smallest composite members are

$$25 = 5^2, \quad 35 = 5 \cdot 7, \quad 49 = 7^2, \quad 55 = 5 \cdot 11, \quad 65 = 5 \cdot 13.$$

Proof. Containment of primes and of their products follows from the previous two propositions. Conversely, if $n \in \mathcal{B}$ then no prime factor of n is 2 or 3, so every prime factor exceeds 3. $\square$

remark 8 (What the branch does and does not decide). Membership in $\mathcal{B}$ is decided by a single residue computation and is therefore not a primality criterion: 35 and 11 both lie on $\mathcal{B}^-$, and 25 and 37 both lie on $\mathcal{B}^+$. The branch records the residue of n modulo 6, an additive datum, whereas primality is multiplicative. This is the same separation formalised in Lemma 1 and Corollary 4.

11.4 Branch transitions under the descent

Proposition 17. Let $n = 6k + \varepsilon$ with $\varepsilon = \pm 1$. The branch of n is determined by ε alone and is independent of the branch of k ; indeed k need not lie in $\mathcal{B}$ at all, since k may be divisible by 2 or 3.

Example 5. $41 = 6 \cdot 7 - 1$ has $k = 7 \in \mathcal{B}^+$, while $47 = 6 \cdot 8 - 1$ has $k = 8 \notin \mathcal{B}$. Both lie on $\mathcal{B}^-$.

remark 9. This is why the canonical algorithm strips factors of 2 and 3 from k before recursing: the descent lands in $\mathbb{Z}_{>0}$, not in $\mathcal{B}$, and only the atoms 2 and 3 restore it.

11.5 Primality as a syntactic property

Proposition 18 (Bracketed canonical form). Modify the canonical form so that every prime atom is enclosed in parentheses: $2 \mapsto (2)$, $3 \mapsto (3)$, and $p \mapsto (2 \cdot 3 \cdot k \pm 1)$ for $p > 3$. Then for every $n > 1$,

$$n \text{ is composite} \iff \text{the substring } *)*(\text{ occurs in } \text{rep}(n).$$

Proof. By construction $\text{rep}(n)$ is a product of one bracketed factor per prime in the factorisation of n , joined by $*$. If n is composite there are at least two such factors, so a $*$ occurs between a closing and an opening bracket. If n is prime there is exactly one factor, and the only $*$ characters occur strictly inside its brackets, where every occurrence is preceded by a digit or a closing bracket of a strictly nested subexpression at depth ≥ 1 . Hence no depth-zero $*$ exists. $\square$

remark 10 (Bracketing is required). Without brackets on the atoms 2 and 3 the test fails: 15 would render as $3*(2*3-1)$ and 9 as $3*3$, neither containing $)*$.

remark 11 (Cost). The test is $O(|\text{rep}(n)|) = O(\log n)$ on a string already in hand. It does not reduce the cost of *obtaining* that string, which requires the prime factorisation of n and dominates. The proposition therefore records that primality is syntactically visible in the notation, not that it is cheaply decidable from n .

```
def rep(n):
    if n <= 3:
        return "(" + str(n) + ")"
    p = smallest_factor(n)
    if p != n:
        return rep(p) + "*" + rep(n // p)
    k, s = ((n+1)//6, "-") if (n+1) % 6 == 0 else ((n-1)//6, "+")
    core = "2*3" if k == 1 else f"2*3*{rep(k)}"
    return f"({core}{s}1)"

def is_prime_form(N):
    if N < 2:
        return f"{N} is neither"
    forms = rep(N)
    label = "not prime" if ")*(" in forms else "prime"
    return f"{label} = {forms} = {N}"
```

Proposition 19 (Factor-free variant). Replacing `smallest_factor` by trial division against 2 and 3 alone yields a map rep_{fast} that is well defined, terminating, and correct in value: $\text{rep}_{\text{fast}}(n)$ evaluates to n for every $n \geq 1$, at cost $O(\log n)$. It is neither canonical nor a primality indicator.

Proof. Termination and correctness are as in Theorem 3, with the descent $n \mapsto k \approx n/6$ interleaved with division by 2 and 3. Non-canonicity and the failure of the syntactic test follow from the single example

$$\text{rep}_{\text{fast}}(25) = (2 \cdot 3 \cdot (2) \cdot (2) + 1),$$

which has no depth-zero product yet $25 = 5^2$ is composite. The canonical form is $(2 \cdot 3 - 1) \cdot (2 \cdot 3 - 1)$. $\square$

remark 12. The two variants isolate the cost precisely. The ± 1 descent is $O(\log n)$ and requires no factorisation; the canonical form requires the full factorisation of n ; and the syntactic primality reading is a property of the canonical form alone. No reorganisation transfers the reading to the cheap variant, since by Lemma 1 the descent is coprime to every level below it.

12 Three levels of structural resolution

The expression tree admits three construction rules, differing only in how much multiplicative structure is exposed at each node. All three produce expressions over the atoms $\{1, 2, 3\}$ that evaluate to n ; they differ in cost and in what the shape of the tree reveals.

12.1 Level 1: blind descent

Definition 5 (Blind form). Define rep_{fast} by peeling factors of 2 and 3 and otherwise applying the descent $n \mapsto (n \mp 1)/6$ with the sign forced by $n \bmod 6$.

Proposition 20. rep_{fast} is total, terminating, and value-correct: $\text{parse}(\text{rep}_{\text{fast}}(n)) = n$ for all $n \geq 1$, at cost $O(\log n)$ arithmetic operations. It exposes no multiplicative structure beyond the primes 2 and 3.

Proposition 21 (Bijectivity). rep_{fast} is a bijection between $\mathbb{Z}_{>0}$ and its image. Consequently every function of the blind tree is a function of n , and no rewriting of the tree can expose a factorisation that was not computed.

Example 6 (The shapes coincide).

$$\text{rep}_{\text{fast}}(25) = (2 \cdot 3 \cdot (2) \cdot (2) + 1), \quad \text{rep}_{\text{fast}}(37) = (2 \cdot 3 \cdot (2) \cdot (3) + 1).$$

The two trees are isomorphic as shapes: a +1 node above a binary product of atoms. Yet $25 = 5^2$ and 37 is prime. No rewrite rule keyed on tree shape can distinguish them.

12.2 Level 2: cheap canonicalisation

Definition 6 (Peel set). Fix a finite set $P = \{p_1 < \dots < p_r\}$ of primes and let q denote the least prime not in P . Define can_P by applying, at each node with value n , the following tests in order, and recursing on the results:

1. if n is a perfect square, $n = m \cdot m$;
2. if $n \equiv 5 \pmod{6}$ and $n + 1 = s^2$, then $n = (s - 1)(s + 1)$;
3. if $n \pm 1 = s^m$ for some $m \geq 2$, apply the cyclotomic factorisation of $s^m \mp 1$ (into cyclotomic polynomials $\Phi_d(s)$, $d \mid m$);
4. if $p \mid n$ for some $p \in P$ with $p \neq n$, split $n = p \cdot (n/p)$;
5. otherwise emit $(2 \cdot 3 \cdot \text{can}_P(k) \pm 1)$.

Theorem 5 (Cost and coverage). can_P runs in $O(|P| \log n)$ arithmetic operations and is value-correct for every n . Its output is canonical — that is, the root operation is a product exactly when n is composite — for every n whose least prime factor lies in P , and for every n caught by rules 1–3. The least n for which can_P fails to be canonical is q^2 .

Proof. Each test is $O(\log n)$: an integer square root, a bounded number of m -th roots, and $|P|$ divisions. If the least prime factor p of n lies in P , rule 4 fires and the root becomes a product. If n is composite with least prime factor $p \notin P$, then $n \geq p^2 \geq q^2$, and rules 1–4 all fail for $n = q^2$ only if q^2 is not a perfect square — which is false, so the first genuine failure is the least composite with least factor q that is not a perfect power, namely $q \cdot q'$ for the next prime q' . Taking $n = q^2$ as the bound is therefore conservative and holds a fortiori. $\square$

corollary 7 (Extensibility). Adjoining a prime p to P costs one further division per node and moves the first structural failure from $q q'$ to the corresponding product for the new least omitted prime. Coverage therefore increases monotonically with $|P|$ and may be pushed to any desired bound B by taking $P = \{p : p \leq \sqrt{B}\}$, at cost $O(\sqrt{B}/\log B)$ divisions per node.

Table 1: Coverage as the peel set grows.

Peel set P	least omitted prime q	first structural failure
$\{2, 3\}$	5	$5 \cdot 7 = 35$
$\{2, 3, 5\}$	7	$7 \cdot 11 = 77$
$\{2, 3, 5, 7, 11, 13\}$	17	$17 \cdot 19 = 323$
$\{2, 3, 5, 7, 11, 13, 17, 19\}$	23	$23 \cdot 29 = 667$
$\{2, \dots, 31\}$	37	$37 \cdot 41 = 1517$
$\{2, \dots, p_r\}$	q	$q q'$

remark 13 (Rules 1–3 are unbounded). Unlike rule 4, the first three tests are not tied to a finite set: they catch every perfect square, every twin product $(s-1)(s+1)$, and every splitting $s^m \pm 1$, at all magnitudes, in $O(\log n)$ time. Their density is thin — perfect powers are sparse — but they contribute at every scale, whereas the peel set contributes only below q^2 .

12.3 Level 3: full canonical form

Proposition 22. Taking P to be all primes up to $\sqrt{n}$ recovers the canonical form of Theorem 1. The cost is that of trial division, $O(\sqrt{n})$, or $O(n^{1/4})$ with Pollard’s rho substituted for rule 4.

12.4 Summary

Table 2: The three levels.

Level	Cost	Value	Structure
rep_{fast}	$O(\log n)$	exact	none beyond 2, 3
can_P	$O(P \log n)$	exact	exact below q^2
rep	$O(\sqrt{n})$	exact	exact

remark 14 (Why the levels do not collapse). The three functions agree on value and disagree only on structure, and the disagreement is not removable by post-processing. By Proposition 21 the blind tree determines n and nothing more, and by Example 6 it does not determine primality even locally. Each additional prime admitted to P purchases a bounded increment of structural resolution at a bounded increment of cost; the limit of this process is trial division. In this sense the three levels are not three algorithms but one algorithm at three settings of a single parameter, and the notation makes the trade-off between cost and resolution explicit rather than implicit.

12.5 Implementation

```

from math import isqrt

def rep_fast(n):
    if n <= 3:
        return "(" + str(n) + ")"
    for d in (2, 3):
        if n % d == 0:
            return f"({d})*" + rep_fast(n // d)
    k, s = ((n+1)//6, "-") if (n+1) % 6 == 0 else ((n-1)//6, "+")

```

```

core = "2*3" if k == 1 else f"2*3*{rep_fast(k)}"
return f"({core}){s}1)"

def can_P(n, P=(2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31)):
    if n <= 3:
        return "(" + str(n) + ")"
    r = isqrt(n)
    if r * r == n:
        h = can_P(r, P)
        return h + "*" + h
    if n % 6 == 5:
        s = isqrt(n + 1)
        if s * s == n + 1:
            return can_P(s - 1, P) + "*" + can_P(s + 1, P)
    for d in P:
        if n != d and n % d == 0:
            return can_P(d, P) + "*" + can_P(n // d, P)
    k, s = ((n+1)//6, "-") if (n+1) % 6 == 0 else ((n-1)//6, "+")
    core = "2*3" if k == 1 else f"2*3*{can_P(k, P)}"
    return f"({core}){s}1)"

def smallest_factor(n):
    for d in (2, 3):
        if n % d == 0:
            return d
    k = 1
    while True:
        for d in (6*k - 1, 6*k + 1):
            if d * d > n:
                return n
            if n % d == 0:
                return d
        k += 1

def rep(n):
    if n <= 3:
        return "(" + str(n) + ")"
    p = smallest_factor(n)
    if p != n:
        return rep(p) + "*" + rep(n // p)
    k, s = ((n+1)//6, "-") if (n+1) % 6 == 0 else ((n-1)//6, "+")
    core = "2*3" if k == 1 else f"2*3*{rep(k)}"
    return f"({core}){s}1)"

```

Proposition 23 (Agreement). For all n , $\text{parse}(f(n)) = n$ where f is any of the three functions. Moreover $\text{can}_P(n) = \text{rep}(n)$ whenever the least prime factor of n lies in P or n is caught by rules 1–3, and $\text{can}_\emptyset(n) = \text{rep}_{\text{fast}}(n)$ when rules 1–3 are also disabled.

13 Generating Pythagorean partners from a fixed leg

13.1 The governing equations

Fix a positive integer A and seek all $B > A$ and $C > 0$ with

$$B^2 - A^2 = C^2. \quad (1)$$

Writing $B = A + d$ for the gap $d = B - A$ and applying the gap identity of Section 2,

$$B^2 - A^2 = 2Ad + d^2 = d(2A + d), \quad (2)$$

so (1) is equivalent to requiring $d(2A + d)$ to be a perfect square. Equivalently, factoring the left side of (1) as a difference of two squares in the other pair of variables,

$$(B - C)(B + C) = A^2. \quad (3)$$

Theorem 6 (Partner enumeration). The solutions (B, C) of (1) in positive integers are in bijection with the factorisations

$$A^2 = de, \quad d < e, \quad d \equiv e \pmod{2},$$

the correspondence being

$$B = \frac{d+e}{2}, \quad C = \frac{e-d}{2}, \quad (4)$$

with inverse $d = B - C$, $e = B + C$.

Proof. Given such a factorisation, (4) produces integers because $d \equiv e \pmod{2}$, and

$$B^2 - C^2 = \left(\frac{d+e}{2}\right)^2 - \left(\frac{e-d}{2}\right)^2 = de = A^2,$$

so (1) holds; $C > 0$ follows from $d < e$. Conversely, given a solution, set $d = B - C$ and $e = B + C$. Then $de = A^2$ by (3), $d < e$ since $C > 0$, and $d + e = 2B$ is even, so $d \equiv e \pmod{2}$. The two maps are mutually inverse. $\square$

corollary 8 (Parity). If A is odd then A^2 is odd, so d and e are both odd and the parity condition is automatic. If A is even then $4 \mid A^2$ and the admissible factorisations are those with both parts even.

corollary 9 (Count). The number of partners of A equals the number of admissible factorisations of A^2 , hence is governed by $\tau(A^2)$. In particular, if A is an odd prime then $A^2 = 1 \cdot A^2$ is the only admissible factorisation and A has the unique partner

$$B = \frac{A^2 + 1}{2}, \quad C = \frac{A^2 - 1}{2}.$$

Example 7. `partners(11)` returns $\{(61, 60)\}$, since 11 is prime; whereas `partners(15)` returns $\{(113, 112), (39, 36), (25, 20), (17, 8)\}$, corresponding to the factorisations $225 = 1 \cdot 225 = 3 \cdot 75 = 5 \cdot 45 = 9 \cdot 25$.

remark 15. Corollary 9 shows that the partner count of A is a disguised divisor count: the multiplicative structure of A is read off directly from the number of right triangles admitting A as a leg.

13.2 Conversion to sums of two squares

Proposition 24. For integers p, q, n ,

$$p^2 + q^2 = n^2 \iff (p - q)^2 + (p + q)^2 = 2n^2.$$

Proof. Expand: $(p - q)^2 + (p + q)^2 = 2p^2 + 2q^2$. □

Since Theorem 6 enumerates triples by leg, and Proposition 24 converts a triple with hypotenuse n into a representation $2n^2 = x^2 + y^2$, iterating **partners** over all legs $\leq N$ and indexing the results by the hypotenuse B yields, for every $n \leq N$, the complete list of such representations. Centres admitting several representations are exactly those n divisible by sufficiently many primes congruent to 1 mod 4.

13.3 Implementation

```
def partners(A):
    """All (B, C) with B^2 - A^2 = C^2."""
    out = []
    n = A * A
    d = 1
    while d * d < n:
        if n % d == 0:
            e = n // d
            if (d + e) % 2 == 0:                # both same parity
                B, C = (d + e) // 2, (e - d) // 2
                if C > 0:
                    out.append((B, C))
            d += 1
    return out
```

The loop condition $d^2 < n$ enumerates the smaller part of each factorisation of $n = A^2$, so each pair $\{d, e\}$ is visited once and $d < e$ holds throughout; the parity test implements the congruence of Theorem 6, and the assignments are (4) verbatim. The cost is $O(A)$ divisions, since d ranges up to $\sqrt{A^2} = A$.

14 Background: the mod-6 wheel

Every integer occupies one of six lanes modulo 6. Lanes 0, 2, 4 are even and lane 3 is divisible by 3, so beyond the base primes 2 and 3 only lanes 1 and 5 can hold primes:

$$p > 3 \implies p \equiv 1 \text{ or } 5 \pmod{6}, \quad \text{i.e.} \quad p = 6m \pm 1.$$

These two survivors are exactly the fold (reset) positions of the two sawtooth waves

$$B = ((x - \frac{6}{8}) \bmod 6) \cdot 2, \quad D = ((x + 2 - \frac{6}{8}) \bmod 6) \cdot 2,$$

which reset at $x \bmod 6 = 1$ and $x \bmod 6 = 5$ respectively.

15 Base-60 inheritance

Base 60 factors as $60 = 6 \times 10$: a mod-6 “tens” wheel geared to a base-10 “units” wheel. We compute in base 10 but time and angle keep the base-60 rollover, whose reset is precisely the $x \bmod 6 = 5$ event of the wheel above. The sieving power of 60 lives entirely in its squarefree kernel $30 = 2 \cdot 3 \cdot 5$.

Details: The Totient Relation

Euler’s totient φ is the hinge of this entire construction. It appears in *two* distinct roles: once attached to the wheel modulus M , and once attached to the integer N under test.

Definition 7 (Totient). For a positive integer n ,

$$\varphi(n) = |\{k : 1 \leq k \leq n, \gcd(k, n) = 1\}|,$$

the count of integers up to n that are coprime to n .

Role 1 — $\varphi(M)$ is the wheel width

For a wheel with modulus M , the spokes are exactly the residues coprime to M , so the number of spokes is $\varphi(M)$ and the surviving density of the number line is $\varphi(M)/M$. This density is governed by the multiplicative product formula.

Theorem 7 (Euler product formula).

$$\varphi(M) = M \prod_{p|M} \left(1 - \frac{1}{p}\right),$$

the product ranging over the *distinct* primes dividing M . Consequently the wheel density is

$$\frac{\varphi(M)}{M} = \prod_{p|M} \left(1 - \frac{1}{p}\right).$$

Each factor $(1 - 1/p)$ is one prime’s worth of lanes struck out. This one equation explains every empirical observation about the wheels:

M	$\varphi(M)$	$\varphi(M)/M$	$\prod(1 - 1/p)$	distinct primes of M
6	2	0.3333	0.3333	$\{2, 3\}$
30	8	0.2667	0.2667	$\{2, 3, 5\}$
60	16	0.2667	0.2667	$\{2, 3, 5\}$ (no gain over 30)
210	48	0.2286	0.2286	$\{2, 3, 5, 7\}$
2310	480	0.2078	0.2078	$\{2, 3, 5, 7, 11\}$

Two consequences follow immediately:

- **60 gains nothing over 30:** they share the distinct primes $\{2, 3, 5\}$, so the product—and hence the density—is identical. The extra factor of 2 in $60 = 2^2 \cdot 3 \cdot 5$ never enters the product, which counts only *distinct* primes.
- **210 does gain:** it introduces a genuinely new factor $(1 - 1/7)$, lowering the density from 0.2667 to 0.2286.

Role 2 — $\varphi(N)$ is a primality signature

For the integer N passed to the scan, the totient encodes primality directly:

$$N \text{ is prime} \iff \varphi(N) = N - 1, \quad \varphi(p^k) = p^k - p^{k-1}.$$

A prime shares a factor with nothing below it, so all $N - 1$ smaller integers are coprime—the maximal possible totient. Every composite satisfies $\varphi(N) < N - 1$.

The bridge — Euler’s theorem

The two roles are joined by Euler’s theorem, which also underlies the fast primality tests (Fermat, Miller–Rabin) that outperform trial division for large N .

Theorem 8 (Euler). If $\gcd(a, N) = 1$ then

$$a^{\varphi(N)} \equiv 1 \pmod{N}.$$

For prime p this specialises, via $\varphi(p) = p - 1$, to Fermat’s little theorem $a^{p-1} \equiv 1 \pmod{p}$.

Miller–Rabin checks whether N *violates* this congruence: a violation certifies compositeness with a single modular exponentiation, whereas the wheel must trial-divide up to $\sqrt{N}$.

Summary

$\underbrace{\varphi(M)}_{\text{sieve width}} = M \prod_{p M} \left(1 - \frac{1}{p}\right) \quad \text{and} \quad \underbrace{\varphi(N) = N - 1}_{\text{primality signature}} \iff N \text{ prime},$

the same function serving once as a wheel’s width and once as an integer’s primality signature, bridged by $a^{\varphi(N)} \equiv 1 \pmod{N}$.

16 Reference implementation

The mod-210 wheel ($210 = 2 \cdot 3 \cdot 5 \cdot 7$, $\varphi(210) = 48$ spokes) used as a trial-division primality scanner:

```
from math import gcd, isqrt

WHEEL210 = [r for r in range(210) if gcd(r, 210) == 1]    # 48 spokes

def wheel_scan_210(N):
    """Smallest factor of N via a mod-210 wheel, or None if N is prime."""
    if N < 2:
        return None
    for p in (2, 3, 5, 7):                                # base primes the wheel skips
        if N == p: return None
        if N % p == 0: return p
    limit = isqrt(N)
    m = 0
    while 210*m + 1 <= limit:
        for r in WHEEL210:
            n = 210*m + r                                # actual candidate = block offset +
            spoke                                     spoke
            if n == 1:                                   # skip trivial divisor
```

```

        continue
    if n > limit:
        break
    if N % n == 0:
        return n                # real factor -> composite
    m += 1
    return None                 # nothing up to sqrt(N) -> prime

def is_prime(N):
    return N >= 2 and wheel_scan_210(N) is None

```

17 The period-six sawtooth and its fold

The trichotomy modulo six can be read dynamically. Writing $6/8 = 0.75$, define the two phase-shifted waves

$$B(x) = ((x - \frac{6}{8}) \bmod 6) \cdot 2, \quad D(x) = ((x + 2 - \frac{6}{8}) \bmod 6) \cdot 2.$$

Proposition 25 (Shape and phase). Both B and D are sawtooths of period 6, and $D(x) = B(x+2)$; that is, D is B advanced by two steps. On integer arguments each takes the half-integer values $\{0.5, 2.5, 4.5, 6.5, 8.5, 10.5\}$, never landing on an integer.

Proof. Periodicity and the identity $D(x) = B(x+2)$ are immediate from the definitions, since $x+2-\frac{6}{8} = (x+2)-\frac{6}{8}$. For integer x the argument $x-\frac{6}{8}$ has fractional part 0.25, so $(x-\frac{6}{8}) \bmod 6 \in \{0.25, 1.25, \dots, 5.25\}$ and multiplication by 2 gives the stated half-integers. The offset $\frac{6}{8} = \frac{3}{4}$ keeps every output off the integer grid, so the wrap point is unambiguous. $\square$

Proposition 26 (The fold locates the residue). D attains its minimum exactly when $x \equiv 5 \pmod{6}$, and B attains its minimum exactly when $x \equiv 1 \pmod{6}$. Equivalently

$$x \equiv 5 \pmod{6} \iff D(x) = 0.5, \quad x \equiv 1 \pmod{6} \iff B(x) = 0.5.$$

The reset (maximum $\rightarrow$ minimum) of each wave marks the residue where it folds. Because $D = B(\cdot + 2)$, the two folds sit at the residues 1 and 5 — precisely the two lanes coprime to six of the Trichotomy proposition, and hence the only lanes a prime > 3 may occupy.

$x \bmod 6$	B	D	note
0	10.5	2.5	
1	0.5	4.5	B resets (prime lane)
2	2.5	6.5	
3	4.5	8.5	
4	6.5	10.5	D peaks
5	8.5	0.5	D resets = $x \equiv 5 \pmod{6}$ (prime lane)

18 Base-sixty inheritance

The wheel of six is inherited by the sexagesimal systems still used for time and angle. The factorisation

$$60 = 6 \times 10$$

gears a mod-6 “tens” wheel to a base-10 “units” wheel: a value v with $0 \leq v < 60$ decomposes as

$$v = 10t + u, \quad t = \left\lfloor \frac{v}{10} \right\rfloor \in \{0, \dots, 5\}, \quad u = v \bmod 10 \in \{0, \dots, 9\}.$$

$t \backslash u$	0	1	2	3	4	5	6	7	8	9
0	0	1	2	3	4	5	6	7	8	9
1	10	11	12	13	14	15	16	17	18	19
2	20	21	22	23	24	25	26	27	28	29
3	30	31	32	33	34	35	36	37	38	39
4	40	41	42	43	44	45	46	47	48	49
5	50	51	52	53	54	55	56	57	58	59

remark 16 (The fold at one half, and the effect of the zero). The tens wheel is capped at 5, the midpoint $5/10 = \frac{1}{2}$ of a decimal digit; its top half $\{6, 7, 8, 9\}$ is never used. This “fold at one half” is an artefact of zero-based indexing: the wheel *counts* six states $\{0, \dots, 5\}$ but *folds* at position 5, the usual gap between an index and a count. If the system omits zero and runs $1, \dots, 6$ instead, index and count coincide: the value is $((x - 1) \bmod 6) + 1$, the fold moves to 6, and the reset condition changes from $x \equiv 5$ to $x \equiv 0 \pmod{6}$ (the multiples of six). The sieving power of 60 resides entirely in its squarefree kernel $30 = 2 \cdot 3 \cdot 5$; the extra factor of 2 in $60 = 2^2 \cdot 3 \cdot 5$ adds no new prime.

19 Wheel factorisation and the totient density

The $6k \pm 1$ enumeration is the wheel of modulus 6. Replacing 6 by a general modulus M and keeping only the residues coprime to M yields a wheel with $\varphi(M)$ spokes, where φ is Euler’s totient. The modulus $210 = 2 \cdot 3 \cdot 5 \cdot 7$ gives $\varphi(210) = 48$ spokes and, used as a trial-division scanner, tests N against the base primes $\{2, 3, 5, 7\}$ and then the wheel candidates up to $\sqrt{N}$.

```
from math import gcd, isqrt

WHEEL210 = [r for r in range(210) if gcd(r, 210) == 1]    # 48 spokes

def wheel_scan_210(N):
    """Smallest factor of N via a mod-210 wheel, or None if N is prime."""
    if N < 2:
        return None
    for p in (2, 3, 5, 7):
        if N == p: return None
        if N % p == 0: return p
    limit = isqrt(N)
    m = 0
    while 210*m + 1 <= limit:
        for r in WHEEL210:
            n = 210*m + r
            if n == 1:
                continue
            if n > limit:
                break
            if N % n == 0:
                return n
        m += 1
```

```

    return None                                     # nothing up to sqrt(N) -> prime

def is_prime(N):
    return N >= 2 and wheel_scan_210(N) is None

```

The three critical corrections over a naive version are: dividing by the actual candidate $n = Mm + r$ rather than the residue r ; skipping $n = 1$, which would otherwise divide every N ; and testing the base primes separately, since the wheel omits them. The scan runs in $O(\sqrt{N})$ divisions scaled by the survivor density $\varphi(M)/M$.

20 The totient relation

Euler's totient φ enters this construction in two distinct roles: once attached to the wheel modulus M , and once attached to the integer N under test.

Definition 8 (Totient). For a positive integer n , $\varphi(n) = |\{k : 1 \leq k \leq n, \gcd(k, n) = 1\}|$, the count of integers up to n coprime to n .

Theorem 9 (Sieve width). A wheel of modulus M has exactly $\varphi(M)$ spokes, and the fraction of the integers it retains is

$$\frac{\varphi(M)}{M} = \prod_{p|M} \left(1 - \frac{1}{p}\right),$$

the product over the distinct primes dividing M .

Each factor $(1 - 1/p)$ removes one prime's lanes. The formula explains the observed densities:

M	$\varphi(M)$	$\varphi(M)/M$	$\prod(1 - 1/p)$	distinct primes
6	2	0.3333	0.3333	$\{2, 3\}$
30	8	0.2667	0.2667	$\{2, 3, 5\}$
60	16	0.2667	0.2667	$\{2, 3, 5\}$ (no gain over 30)
210	48	0.2286	0.2286	$\{2, 3, 5, 7\}$
2310	480	0.2078	0.2078	$\{2, 3, 5, 7, 11\}$

Thus 60 yields no improvement over 30 (identical distinct primes), whereas 210 tightens the sieve by folding in the new factor $(1 - 1/7)$.

Proposition 27 (Primality signature). For the integer N under test, N is prime iff $\varphi(N) = N - 1$, and $\varphi(p^k) = p^k - p^{k-1}$.

Theorem 10 (Euler, the bridge). If $\gcd(a, N) = 1$ then $a^{\varphi(N)} \equiv 1 \pmod{N}$. For prime p this reduces, via $\varphi(p) = p - 1$, to Fermat's little theorem $a^{p-1} \equiv 1 \pmod{p}$.

The two roles are the same function seen twice: $\varphi(M)$ as the width of the sieve (Theorem 9) and $\varphi(N) = N - 1$ as the signature of primality, joined by Euler's congruence, which is the basis of the fast tests (Fermat, Miller–Rabin) that supersede trial division for large N .

21 Details: reciprocity of the sieve factor and the base- p expansion

The wheel multiplies the factors $(1 - 1/p)$; the complementary object is their reciprocal, the base- p geometric series. The two are related by a single inverse identity, which the following floating-point experiment exhibits in base 2.

```
for x in range(2, 100):
    A = ((x + (1/x)) % x)
    if (x*A % x) == 1.0:
        print(x, (x*A % x))
# printed: 2, 4, 8, 16, 32, 64
```

Proposition 28 (What the loop computes). For every integer $x \geq 2$ one has $A = (x+1/x) \bmod x = 1/x$, hence $x \cdot A = 1$ and $(x \cdot A) \bmod x = 1$. In exact arithmetic the test $x \cdot A = 1$ therefore holds for *all* x : it is the multiplicative inverse identity $x \cdot (1/x) = 1$.

remark 17 (Why only the powers of two are printed). The equality is with the literal 1.0 in IEEE-754 double precision, which is base 2. The reciprocal $1/x$ is represented *exactly* only when $x = 2^J$, since then $1/x = 2^{-J}$ is a finite binary fraction; the round-trip $(x+1/x) \bmod x$ then returns exactly 2^{-J} and $x \cdot A$ is exactly 1.0. For any other x the product rounds to $1.0 \pm \varepsilon$ and fails the test. Thus the loop is, in effect, a detector of the powers of two — the base-2 signature.

Proposition 29 (Sieve factor and base- p series are reciprocal). For every prime p ,

$$\frac{1}{1 - \frac{1}{p}} = \sum_{J \geq 0} \frac{1}{p^J} = 1 + \frac{1}{p} + \frac{1}{p^2} + \cdots, \quad \text{so} \quad \left(1 - \frac{1}{p}\right) \cdot \sum_{J \geq 0} \frac{1}{p^J} = 1.$$

This is the local Euler factor of the Riemann product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ at $s = 1$. The sieve factor $(1 - 1/p)$ measures what *survives* the wheel (the lanes coprime to p); its reciprocal $\sum_J p^{-J}$ *generates* what the wheel removes (the powers and multiples of p). Their product is 1 — the same inverse identity $x \cdot (1/x) = 1$ that the loop tests.

p	$1 - 1/p$ (sieve)	$1/(1 - 1/p) = \sum_J p^{-J}$	product
2	0.5000	$2.0000 = \sum_J 2^{-J}$	1
3	0.6667	1.5000	1
5	0.8000	1.2500	1
7	0.8571	1.1667	1

For $p = 2$ the reciprocal series is the geometric expansion

$$\frac{1}{1 - \frac{1}{2}} = \sum_{J \geq 0} \frac{1}{2^J} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 2,$$

the exact 2^J series of the loop and the reciprocal of the wheel's smallest factor $(1 - \frac{1}{2})$. Because floating-point arithmetic is itself base 2, this is the one Euler factor the machine represents exactly, which is why the inverse test $x \cdot (1/x) = 1$ selects precisely the powers of two. In summary,

$$\underbrace{\prod_p \left(1 - \frac{1}{p}\right)}_{\text{wheel: keeps coprimes}} \quad \text{and} \quad \underbrace{\prod_p \frac{1}{1 - \frac{1}{p}} = \prod_p \sum_{J \geq 0} p^{-J}}_{\text{base-}p \text{ generator}}$$

are reciprocal expressions of the same primes, meeting at $x \cdot (1/x) = 1$.

22 The square-root staircase

Pairing the inverse identity of the previous section with the integer square root exposes a second structure: a step function whose runs are the odd numbers.

```
from math import isqrt
for x in range(3, 100):
    A = ((x + 1/x) % x)      # = 1/x    -> x*A % x approx 1.0
    x1 = isqrt(x)            # the "category" floor(sqrt(x))
    A1 = ((x1 + 1/x1) % x1)  # = 1/x1   -> x1*A1 % x1 approx 1.0
    print(x, (x*A % x), x1, (x1*A1 % x1))
```

The two remainder columns are both the constant ≈ 1 of the identity $x \cdot (1/x) = 1$; the information is carried entirely by $x_1 = \lfloor \sqrt{x} \rfloor$, a staircase that is flat on each block and steps up by one at every perfect square.

Proposition 30 (Odd-length runs). Fix $k \geq 1$. Then $\lfloor \sqrt{x} \rfloor = k$ precisely for $k^2 \leq x \leq (k+1)^2 - 1$, a run of length

$$(k+1)^2 - k^2 = 2k + 1.$$

Consequently the gaps between consecutive squares are the odd numbers, and

$$k^2 = 1 + 3 + 5 + \cdots + (2k-1) = \sum_{j=1}^k (2j-1).$$

Proof. $\lfloor \sqrt{x} \rfloor = k \iff k \leq \sqrt{x} < k+1 \iff k^2 \leq x < (k+1)^2$, so the run is the integer interval $[k^2, (k+1)^2 - 1]$ of length $(k+1)^2 - k^2 = 2k+1$. Summing these increments telescopes: $\sum_{j=1}^k (2j-1) = k^2$. $\square$

$k = \lfloor \sqrt{x} \rfloor$	x range	run length	$(k+1)^2 - k^2$
4	16...24	9	9
5	25...35	11	11
6	36...48	13	13
7	49...63	15	15
8	64...80	17	17

The category k persists for exactly $2k+1$ consecutive integers, turning over only at the boundary squares k^2 . The square numbers are the partial sums of the odd numbers, and $\lfloor \sqrt{x} \rfloor$ is the step function counting which shell x occupies.

22.1 Connection to the divisor bound and to differences of squares

The staircase links two earlier constructions.

remark 18 (Trial-division bound). In the wheel scan the search is capped at $\text{limit} = \lfloor \sqrt{N} \rfloor$. For all x in one run of the staircase, $\sqrt{x}$ lies in the same integer shell $[k, k+1)$; the “category” is precisely the divisor bound below which a factor, if any, must appear.

remark 19 (Fermat descent walks the staircase). The difference-of-squares identity $n = a^2 - b^2 = (a-b)(a+b)$ is searched by starting at $a = \lfloor \sqrt{n} \rfloor$ and increasing a , testing whether $a^2 - n$ is a perfect square. Each increment raises a^2 by the odd number $2a+1$, so Fermat’s method advances exactly along the square staircase of the proposition; the odd-gap law is its step size.

22.2 Higher power staircases

Nesting the integer square root raises the staircase one level up the power ladder. Since the nested floors collapse,

$$\lfloor \sqrt{\lfloor \sqrt{x} \rfloor} \rfloor = \lfloor x^{1/4} \rfloor,$$

the double root `isqrt(isqrt(x))` is the integer fourth root, constant between consecutive fourth powers and stepping at 16, 81, 256, 625, ...

```
from math import isqrt
for x in range(4, 100):
    A = ((x + 1/x) % x)
    x1 = isqrt(isqrt(x))          # integer 4th root: floor(x**(1/4))
    A1 = ((x1 + 1/x1) % x1)
    print(x, (x*A % x), x1, (x1*A1 % x1))
```

Proposition 31 (Fourth-power runs). $\lfloor x^{1/4} \rfloor = k$ precisely for $k^4 \leq x \leq (k+1)^4 - 1$, a run of length

$$(k+1)^4 - k^4 = 4k^3 + 6k^2 + 4k + 1.$$

$k = \lfloor x^{1/4} \rfloor$	x range	run length	$(k+1)^4 - k^4$
1	1 ... 15	15	15
2	16 ... 80	65	65
3	81 ... 255	175	175

The run for $k = 2$ is exactly $3^4 - 2^4 = 81 - 16 = 65$.

Theorem 11 (Binomial gap law). For each integer $m \geq 1$, the map $x \mapsto \lfloor x^{1/m} \rfloor$ is constant on $[k^m, (k+1)^m - 1]$, and the length of that run is the discrete derivative of k^m :

$$(k+1)^m - k^m = \sum_{j=0}^{m-1} \binom{m}{j} k^j.$$

The coefficients are row m of Pascal's triangle with the leading k^m term removed.

Proof. $\lfloor x^{1/m} \rfloor = k \iff k^m \leq x < (k+1)^m$, giving the stated interval and length $(k+1)^m - k^m$. The binomial theorem expands $(k+1)^m = \sum_{j=0}^m \binom{m}{j} k^j$; subtracting the $j = m$ term k^m leaves $\sum_{j=0}^{m-1} \binom{m}{j} k^j$. $\square$

The earlier square-root staircase is the case $m = 2$, whose gap $2k+1$ is the odd numbers (Pascal row 2); the fourth root is $m = 4$, gap $4k^3 + 6k^2 + 4k + 1$ (row 4: 1, 4, 6, 4, 1). Each further nesting of `isqrt` doubles the exponent:

$$m = 2 : 2k+1, \quad m = 3 : 3k^2+3k+1, \quad m = 4 : 4k^3+6k^2+4k+1, \quad m = 5 : 5k^4+10k^3+10k^2+5k+1.$$

Throughout, the reciprocal columns remain the invariant $x \cdot (1/x) = 1$; only the root-category x_1 advances up the power ladder, its run lengths given by Theorem 11.

22.3 Stepping by more than one

Replacing the unit step by an arbitrary step d generalises the gap law of Theorem 11: the $+1$ becomes a $+d$, and each binomial coefficient picks up a power of d .

Theorem 12 (Step- d gap law). For integers $m \geq 1$ and $d \geq 1$,

$$(k+d)^m - k^m = \sum_{j=0}^{m-1} \binom{m}{j} d^{m-j} k^j.$$

The case $d = 1$ recovers Theorem 11.

Proof. By the binomial theorem $(k+d)^m = \sum_{j=0}^m \binom{m}{j} d^{m-j} k^j$; the $j = m$ term is $d^0 k^m = k^m$, and subtracting it leaves the stated sum. $\square$

For $d = 2$ the coefficients are $\binom{m}{j} 2^{m-j}$:

$$m = 2 : 4k+4, \quad m = 3 : 6k^2+12k+8, \quad m = 4 : 8k^3+24k^2+32k+16, \quad m = 5 : 10k^4+40k^3+80k^2+80k+32.$$

Proposition 32 (Telescoping). The step- d gap is the sum of d consecutive unit gaps,

$$(k+d)^m - k^m = \sum_{i=0}^{d-1} [(k+i+1)^m - (k+i)^m],$$

so $(k+2)^m - k^m$ is the combined length of the two adjacent shells $\lfloor x^{1/m} \rfloor \in \{k, k+1\}$, i.e. the run $k^m \leq x \leq (k+2)^m - 1$. For example at $m = 4$, $k = 2$ this is $65 + 175 = 240$.

remark 20 (Divisibility and difference of squares). Since $a^m - b^m$ is divisible by $a - b$, the step- d gap is always a multiple of d . In particular for $m = 2$ it is a pure difference of squares,

$$(k+d)^2 - k^2 = (k+d-k)(k+d+k) = d(2k+d),$$

and with $d = 2$, $(k+2)^2 - k^2 = 4(k+1)$. This returns the staircase gap to the difference-of-squares identity of Section 2, linking the power-shell runs to the factorisation method of the earlier sections.

23 When the gap is a square: Pythagorean triples

Specialising the step- d gap law to $m = 2$ and factoring the difference of squares gives

$$(k+d)^2 - k^2 = (k+d-k)(k+d+k) = d(2k+d),$$

so the square $(k+d)^2$ splits as

$$\boxed{k^2 + d(2k+d) = (k+d)^2}.$$

Theorem 13 (Square gap $\Rightarrow$ triple). If the gap $d(2k+d)$ is itself a perfect square, say $d(2k+d) = s^2$, then

$$k^2 + s^2 = (k+d)^2,$$

i.e. $(k, s, k+d)$ is a Pythagorean triple with legs k and s and hypotenuse $k+d$. Conversely every triple arises this way, with d the excess of the hypotenuse over the leg k .

Proof. Substituting $d(2k+d) = s^2$ into $k^2 + d(2k+d) = (k+d)^2$ gives $k^2 + s^2 = (k+d)^2$. Conversely, given $k^2 + s^2 = h^2$ set $d = h - k$; then $d(2k+d) = (h-k)(h+k) = h^2 - k^2 = s^2$. $\square$

Thus a Pythagorean triple is exactly a staircase gap $d(2k+d)$ that happens to be a square: the difference of two squares that is again a square. This is the enumeration of Section 3 (Pythagorean partners) read through the gap law — the leg k plays the role of the fixed leg, $k+d$ the hypotenuse.

k	d	$d(2k+d) = s^2$	s	triple $(k, s, k+d)$
3	2	16	4	(3, 4, 5)
4	1	9	3	(4, 3, 5)
5	8	144	12	(5, 12, 13)
8	9	225	15	(8, 15, 17)
7	18	576	24	(7, 24, 25)

remark 21 (Toward squares of squares). Requiring instead that the *two* flanking gaps be equal, $c^2 - b^2 = b^2 - a^2$, yields $a^2 + c^2 = 2b^2$: three squares in arithmetic progression. That is the center law $p^2 + q^2 = 2e^2$ of the 3×3 magic square of squares, where each line through the centre is such a progression. Theorem 13 supplies the single-square (triple) condition; the magic square of squares is the simultaneous, concentric version of it, developed separately.

24 Canonical classification of triples by the $\{3, 4, 5\}$ clusters

Drop the leg/hypotenuse labels and view a triple as a set $\{A, B, C\}$ with $C = \max = k+d$. Classify it by how the divisibility tokens 3, 4, 5 distribute among the three numbers.

Lemma 2 (The maximum carries at most the 5). In a primitive triple, $\max(A, B, C)$ is divisible by neither 3 nor 4; it may be divisible by 5, or by none of the three.

Proof. Write $C = \max$, so $C^2 = A^2 + B^2$. Squares are $\equiv 0, 1 \pmod{3}$; if $3 \mid C$ then $A^2 + B^2 \equiv 0$ forces $3 \mid A$ and $3 \mid B$, contradicting primitivity. And a primitive hypotenuse is odd, so $2 \nmid C$, hence $4 \nmid C$. Only 5 may divide C . $\square$

Theorem 14 (Five clusters). Each of 3, 4, 5 divides exactly one of the three numbers, and by Lemma 2 the tokens 3 and 4 fall on the two smaller numbers. Consequently every primitive triple falls into exactly one of five clustering patterns:

pattern	cluster on $\{A, B, C\}$	example
all separate	$\{3\} \{4\} \{5\}$	(3, 4, 5)
pair $3 \cdot 4 = 12$	$\{12\} \{5\} \{-\}$	(5, 12, 13)
pair $3 \cdot 5 = 15$	$\{15\} \{4\} \{-\}$	(8, 15, 17)
pair $4 \cdot 5 = 20$	$\{20\} \{3\} \{-\}$	(20, 21, 29)
all together = 60	$\{60\} \{-\} \{-\}$	(11, 60, 61)

Here “—” marks a number coprime to 30. In the four patterns containing a pair or the full 60, that cluster is forced onto the two smaller numbers, and the maximum $C = k+d$ carries only the lone 5 (pattern “all separate”) or nothing at all.

corollary 10 (The 60 invariant). Since each token 3, 4, 5 occurs exactly once among A, B, C ,

$$60 = 3 \cdot 4 \cdot 5 \mid A B C$$

for every Pythagorean triple, regardless of which of the five patterns it realises.

remark 22 (Reversal symmetry). Because (A, B, C) and (C, B, A) denote the same triple, sorting $A \leq B \leq C$ is a canonical form; the classification above is stated on the sorted set, with $C = k + d$ the distinguished maximum. The priority order $5 > 4 > 3$ is the natural reading direction: 5 is the mobile token (it selects whether C owns it or not), while 3 and 4 are pinned to the two smaller numbers.

25 The sum rule: every line sums to $3k^2$

The classification of triples feeds directly into the 3×3 magic square of squares. Write the square with centre E ; the center law of that problem forces the magic sum to be $S = 3E$, and each cell paired with its reflection through the centre sums to $2E$.

Definition 9 (Square-centred magic square). Fix k and take the centre cell to be the square $E = k^2$. Then the magic sum is

$$S = 3E = 3k^2,$$

and each pair of opposite cells $\{a^2, b^2\}$ must satisfy $a^2 + b^2 = 2k^2$.

Proposition 33 (The sum rule). With centre k^2 , a line through the centre carries cells a^2, k^2, b^2 , and

$$a^2 + k^2 + b^2 = 3k^2 \iff a^2 + b^2 = 2k^2.$$

Hence assembling the square reduces to finding representations of $2k^2$ as a sum of two squares; every line then sums automatically to $3k^2$.

Proposition 34 (Pairs are hypotenuse- k triples). For $a < b$ with $a^2 + b^2 = 2k^2$, the numbers a, b share parity, and

$$p = \frac{a+b}{2}, \quad q = \frac{b-a}{2} \implies p^2 + q^2 = k^2,$$

so each nontrivial opposite-cell pair corresponds to a Pythagorean triple with *hypotenuse* k . This complements Theorem 13, where k was a leg: here k is the hypotenuse, and the count of pairs grows with the number of prime factors of k congruent to 1 mod 4.

Proof. $a^2 + b^2 = 2k^2$ is even, so $a \equiv b \pmod{2}$ and p, q are integers. Then $p^2 + q^2 = \frac{(a+b)^2 + (b-a)^2}{4} = \frac{2a^2 + 2b^2}{4} = \frac{a^2 + b^2}{2} = k^2$. $\square$

Example 8 (The 425 centre). For $k = 425 = 5^2 \cdot 17$ (both $5 \equiv 17 \equiv 1 \pmod{4}$), the equation $a^2 + b^2 = 2 \cdot 425^2 = 361250$ has eight solutions with $a \leq b$,

$$(7, 601), (85, 595), (175, 575), (205, 565), (289, 527), (329, 503), (355, 485), (425, 425),$$

seven of them nontrivial. Each gives a line summing to $3 \cdot 425^2 = 541875$; the abundance of representations is what makes 425^2 the classical near-miss centre for a magic square of nine squares.

```

from math import isqrt

def pairs_to_2ksq(k):
    """All (a,b), a<=b, with a^2 + b^2 = 2*k^2: the opposite-cell pairs of
    a
    magic square of squares centred at k^2. Each line through the centre
    sums
    to a^2 + k^2 + b^2 = 3*k^2."""
    target, out, a = 2*k*k, [], 0
    while 2*a*a <= target:
        b2 = target - a*a
        b = isqrt(b2)
        if b*b == b2 and b >= a:
            out.append((a, b))
        a += 1
    return out

```

The sum rule $a^2 + b^2 = 2k^2$ is thus the necessary skeleton: it guarantees every line through the centre meets $3k^2$. The remaining difficulty of the magic square of squares — making the three off-centre lines close simultaneously — is the compatibility of several such pairs on one centre, developed in the companion magic-square analysis.

remark 23 (Two sums in the same k, s, d letters). The two summations are easily conflated, so we state both in the *same* variables k, s, d ; they differ only in whether k is a leg or the hypotenuse.

Universal identity (k a leg). For every triple $(k, s, k + d)$ with $k^2 + s^2 = (k + d)^2$,

$$k^2 + s^2 + (k + d)^2 = (k + d)^2 + (k + d)^2 = 2(k + d)^2 \quad (\text{always}),$$

the three squares sum to twice the hypotenuse square automatically: the largest already equals the sum of the other two, so adding it again merely doubles it.

Magic sum (k the hypotenuse). Take instead a triple with $s^2 + d^2 = k^2$. Then the centre k^2 flanked by $(d - s)^2$ and $(d + s)^2$ gives

$$(d - s)^2 + k^2 + (d + s)^2 = 2(s^2 + d^2) + k^2 = 2k^2 + k^2 = 3k^2 \quad (\text{only when } s^2 + d^2 = k^2),$$

since $(d - s)^2 + (d + s)^2 = 2(s^2 + d^2) = 2k^2$. This is the pair $\{a, b\} = \{d - s, d + s\}$ of Proposition 33 written through the legs s, d .

Thus the same three letters give both rules: with k a leg the three squares sum to $2(k + d)^2$ unconditionally, while with k the hypotenuse the balanced line $(d - s)^2, k^2, (d + s)^2$ sums to $3k^2$ — the scarce, conditional sum the magic square of squares demands on every line at once. For $k = 5$, $(s, d) = (3, 4)$ this is the progression 1, 25, 49.

25.1 The magic line is asymmetric: two gaps, not one

The flankers of a magic line do not sit at equal distances from the centre. Write the three roots of a line centred at k^2 as

$$(k - d_1, k, k + d_2),$$

so d_1 is the gap below the centre and d_2 the gap above it.

Proposition 35 (Two-gap condition). The three squares $(k - d_1)^2$, k^2 , $(k + d_2)^2$ form a magic line (sum $3k^2$, equivalently $(k - d_1)^2 + (k + d_2)^2 = 2k^2$) if and only if

$$d_1^2 + d_2^2 = 2k(d_1 - d_2).$$

In particular $d_1 > d_2 > 0$: the line is necessarily *asymmetric*, since $d_1 = d_2$ forces $d_1 = d_2 = 0$.

Proof. Expand $(k - d_1)^2 + (k + d_2)^2 = 2k^2 - 2kd_1 + d_1^2 + 2kd_2 + d_2^2$. Setting this equal to $2k^2$ gives $d_1^2 + d_2^2 = 2k(d_1 - d_2)$. The right side is positive, so $d_1 > d_2$; and $d_1 = d_2$ collapses both to 0. $\square$

The two gaps are supplied by a hypotenuse-triple $s^2 + d^2 = k^2$ through

$$d_1 = k - (d - s), \quad d_2 = (d + s) - k,$$

the flanker roots being $d - s$ and $d + s$. For instance, $k = 442$ with $(s, d) = (42, 440)$ gives roots $(398, 442, 482)$, hence $d_1 = 44$, $d_2 = 40$ and $44^2 + 40^2 = 3536 = 2 \cdot 442 \cdot (44 - 40)$.

remark 24 (Why a leg-triple's single d cannot build a line). Proposition 35 shows a magic line is a *two-parameter* object (d_1, d_2) . A leg-triple $(k, s, k + d)$ from $s^2 = d(2k + d)$ carries only *one* gap d : the quantity $(d - s)^2$ is a perfect square automatically, but the opposite cell $2k^2 - (d - s)^2$ is then forced to use the same single d , pinning the second gap to the first. Since a genuine line needs two independent gaps obeying $d_1^2 + d_2^2 = 2k(d_1 - d_2)$, the value $2k^2 - (d - s)^2$ is in general not a square even though the formal sum still reads $3k^2$. This is exactly why magic lines come from hypotenuse-triples (k the hypotenuse), never from the single- d leg-triples.

25.2 Counting the lines: rectangles on a fixed diagonal

Each magic line is an integer rectangle $a \times b$ (with $a < b$) whose diagonal equals that of the $k \times k$ square, namely $a^2 + b^2 = 2k^2$: a lattice point on the circle $x^2 + y^2 = 2k^2$. Their number is fixed by the factorisation of k .

Theorem 15 (Number of magic lines). Write $k = 2^a \prod_i p_i^{e_i} \prod_j q_j^{f_j}$ with $p_i \equiv 1 \pmod{4}$ and $q_j \equiv 3 \pmod{4}$. The number of magic lines centred at k^2 — equivalently the number of integer rectangles $a \times b$, $a < b$, with diagonal $k\sqrt{2}$ — is

$$N(k) = \frac{1}{2} \left(\prod_{p_i \equiv 1(4)} (2e_i + 1) - 1 \right).$$

Only the primes $p_i \equiv 1 \pmod{4}$ contribute; the factor 2^a and the primes $q_j \equiv 3 \pmod{4}$ add nothing.

The reason is Gaussian: in $\mathbf{Z}[i]$ a prime $\equiv 1 \pmod{4}$ splits, producing new representations as a sum of two squares, whereas 2 ramifies and primes $\equiv 3 \pmod{4}$ stay inert and contribute none. The map $a^2 + b^2 = 2k^2 \Leftrightarrow p^2 + q^2 = k^2$ with $p = \frac{a+b}{2}$, $q = \frac{b-a}{2}$ identifies these rectangles with the representations of k as a hypotenuse, whence the odd-divisor count $\prod (2e_i + 1)$. The formula was checked against direct enumeration for all $k < 3000$.

k	primes $\equiv 1(4)$	$N(k)$
$441 = 3^2 7^2$	none	0
$442 = 2 \cdot 13 \cdot 17$	13, 17	$\frac{3 \cdot 3 - 1}{2} = 4$
$425 = 5^2 \cdot 17$	5^2 , 17	$\frac{5 \cdot 3 - 1}{2} = 7$
$625 = 5^4$	5^4	$\frac{9 - 1}{2} = 4$
$1105 = 5 \cdot 13 \cdot 17$	three	$\frac{2 \cdot 7 - 1}{2} = 13$

corollary 11 (Entry price for a magic square). A 3×3 magic square of squares needs four lines through its centre, hence $N(k) \geq 4$, i.e.

$$\prod_{p_i \equiv 1(4)} (2e_i + 1) \geq 9.$$

The minimal centres therefore carry either two distinct primes $\equiv 1 \pmod{4}$ (as $5 \cdot 13 = 65$) or one such prime to the fourth power (as $5^4 = 625$); the smallest are $k = 65, 85, 130, 145, \dots$

Thus a single prime $\equiv 1 \pmod{4}$ (like $k = 5$, $N = 1$) fixes one rectangle on the diagonal but cannot seat a square; each additional such prime multiplies the rectangle count, and four rectangles on one diagonal $k\sqrt{2}$ is the threshold below which no magic square of squares can exist.

25.3 An area condition for a magic line

The diagonal $k\sqrt{2}$ cuts the rectangle $a \times b$ into two congruent right triangles, each with legs a, b , hypotenuse $k\sqrt{2}$, and area $A/2$, where $A = ab$ is the rectangle area. Fixing the diagonal is $a^2 + b^2 = 2k^2$, so

$$(a + b)^2 = 2k^2 + 2A, \quad (b - a)^2 = 2k^2 - 2A.$$

Proposition 36 (Area existence condition). A magic line of area A exists at centre k^2 if and only if both

$$2(k^2 + A) \quad \text{and} \quad 2(k^2 - A)$$

are perfect squares, in which case $a + b = \sqrt{2(k^2 + A)}$ and $b - a = \sqrt{2(k^2 - A)}$ recover the sides.

Proof. $a^2 + b^2 = 2k^2$ and $2ab = 2A$ give $(a \pm b)^2 = 2k^2 \pm 2A$; integrality of a, b is equivalent to both right-hand sides being perfect squares, and the displayed formulas invert the system. $\square$

Thus one may scan the *area* A and ask that it sit symmetrically between two squares about $2k^2$, rather than scanning a side. For $k = 5$, $A = 7$: $2(25 + 7) = 64 = 8^2$ and $2(25 - 7) = 36 = 6^2$, so $a + b = 8$, $b - a = 6$, the rectangle 1×7 and the line 1, 25, 49.

Proposition 37 (The area carries the congruum). Writing the flanker roots as $a = d - s$, $b = d + s$ from the Pythagorean triangle $s^2 + d^2 = k^2$, the arithmetic progression a^2, k^2, b^2 has common difference

$$2sd = 4 \cdot \left(\frac{1}{2}sd\right) = 4 \cdot (\text{area of the } (s, d, k) \text{ triangle}),$$

and the rectangle area and this congruum are the legs of a further triple,

$$(d^2 - s^2)^2 + (2sd)^2 = (k^2)^2.$$

For $k = 5$ this is (7, 24, 25): rectangle area 7, congruum 24, hypotenuse $k^2 = 25$. The identity is the classical *congruent-number* bridge — the magic-line step is four times the area of a right triangle with hypotenuse k — so the existence of magic lines at k^2 is the existence of such right-triangle areas, again controlled by the primes $\equiv 1 \pmod{4}$ of Theorem 15.

```
from math import isqrt

def is_square(N):
    if N < 0: return False
    r = isqrt(N); return r*r == N
```

```

def lines_by_area(k):
    """Magic lines at centre  $k^2$  found by the AREA condition:
     $A = a*b$  is valid  $\iff 2(k^2 + A)$  and  $2(k^2 - A)$  are both perfect
    squares."""
    out = []
    for A in range(1, k*k + 1):
        if is_square(2*(k*k + A)) and is_square(2*(k*k - A)):
            apb, bma = isqrt(2*(k*k + A)), isqrt(2*(k*k - A))
            a, b = (apb - bma)//2, (apb + bma)//2
            if 0 < a < b:
                out.append((A, a, b))          # rectangle area, sides
    return out

```

25.4 The area search bound and the pair count

Proposition 38 (Area is capped at k^2). Every magic-line area satisfies $A \leq k^2$, with equality only for the degenerate line $a = b = k$. Hence the area search never needs to exceed k^2 .

Proof. The condition of Proposition 36 requires $2(k^2 - A)$ to be a perfect square, so $A \leq k^2$; equivalently, by AM–GM, $A = ab \leq \frac{a^2+b^2}{2} = k^2$, with equality iff $a = b = k$. $\square$

corollary 12 (Pair count and the magic-square threshold). Scanning $1 \leq A \leq k^2$ returns the degenerate line $A = k^2$ together with exactly $N(k)$ nontrivial magic lines, where (Theorem 15)

$$N(k) = \frac{1}{2} \left(\prod_{p|k, p \equiv 1(4)} (2e_p + 1) - 1 \right).$$

A 3×3 magic square of squares therefore requires

$$N(k) \geq 4 \iff \prod_{p \equiv 1(4)} (2e_p + 1) \geq 9,$$

i.e. two distinct primes $\equiv 1 \pmod{4}$, or one such prime to the fourth power.

Thus bounding the area at k^2 discards nothing, and the number of usable pairs is fixed entirely by the primes $\equiv 1 \pmod{4}$ dividing k : the degenerate endpoint $A = k^2$ is dropped, and at least four of the remaining area-pairs are needed before a magic square of squares can exist. The following table shows the search window and the count together.

k	area window $[A_{\min}, k^2]$	nontrivial $N(k)$	square possible?
5	[7, 25]	1	no
65	[1183, 4225]	4	yes (threshold)
425	[4207, 180625]	7	yes
442	[38564, 195364]	4	yes (threshold)

26 Assembling the square: two forced cells

Given several pairs on a centre k^2 , a magic square is not obtained by checking each pair (every pair already sums to $3k^2$: it is a line). One must interlock two pairs. Label the grid

$$\begin{pmatrix} A & B & C \\ D & E & F \\ G & H & I \end{pmatrix}, \quad E = k^2,$$

and take two pairs as the diagonals, so $A + I = C + G = 2k^2$; let the middle-column and middle-row pairs give $B + H = D + F = 2k^2$.

Proposition 39 (Only two independent off-centre conditions). The four lines through the centre sum to $3k^2$ automatically. Among the four off-centre lines only two are independent: since the opposite cells pair to $2k^2$,

$$\text{bottom row} = 6k^2 - \text{top row}, \quad \text{right col} = 6k^2 - \text{left col},$$

so imposing top row $= 3k^2$ forces the bottom row, and left col $= 3k^2$ forces the right column.

corollary 13 (The winning test: two forced cells). Fixing the corners A, C (from the two diagonals, with $I = 2k^2 - A$, $G = 2k^2 - C$), the row and column conditions determine the edges

$$B = 3k^2 - A - C, \quad D = k^2 - A + C$$

(and $H = 2k^2 - B$, $F = 2k^2 - D$). All eight line-sums are then $3k^2$, so the square of squares exists at this centre iff the two *forced* cells B and D are themselves perfect squares.

Thus, whatever the number $N(k)$ of pairs, assembling a square reduces to: pick two diagonal pairs, compute the two forced edges $B = 3k^2 - A - C$ and $D = k^2 - A + C$, and test only those for squareness. For $k = 8840$ ($N = 13$ pairs) no choice of two diagonals yields square B and D ; the square of nine squares remains, in general, an open problem.

```
from math import isqrt
def is_square(N):
    r = isqrt(N); return N >= 0 and r*r == N

def winners(k):
    """Magic squares of squares at centre k^2. Take two pairs as the
    diagonals
    (corners A, C); the edges B = 3k^2 - A - C and D = k^2 - A + C are
    FORCED and
    must also be perfect squares."""
    c = k*k
    corners = [(a*a, 2*c - a*a) for a in range(1, k)
               if is_square(2*c - a*a) and isqrt(2*c - a*a) > k]
    out = []
    for i in range(len(corners)):
        for j in range(len(corners)):
            if i == j: continue
            for A in corners[i]:
                for C in corners[j]:
                    B, D = 3*c - A - C, c - A + C
                    if B > 0 and D > 0 and is_square(B) and is_square(D):
                        out.append((A, C, B, D)) # a full nine-square
                                                    solution
    return out
```

27 The governing quantity $b^2 - a^2$ and its additive-closure rule

For a fixed centre k^2 the diagonal $k\sqrt{2}$ and the number $N(k)$ of rectangles are already determined (Theorem 15). What remains is a relation among the *sides* of those rectangles. The controlling quantity is the difference of the squared sides.

Definition 10 (Eccentricity of a rectangle). For a rectangle with sides $a < b$ on the diagonal $k\sqrt{2}$ (so $a^2 + b^2 = 2k^2$) set

$$\mu = b^2 - a^2 = 2\delta = 4sd,$$

where $\delta = k^2 - a^2$ is its offset and s, d the legs of the associated hypotenuse- k triangle ($a = d - s$, $b = d + s$). Then μ measures how far the rectangle departs from the $k \times k$ square.

Theorem 16 (Additive-closure rule). Two rectangles (a_1, b_1) and (a_2, b_2) , taken as the two diagonals of the square, close into a magic square of squares centred at k^2 if and only if both

$$\mu_1 + \mu_2 = b_3^2 - a_3^2 \quad \text{and} \quad |\mu_1 - \mu_2| = b_4^2 - a_4^2$$

for rectangles 3, 4 in the set; equivalently, the two forced edges

$$3k^2 - a_1^2 - a_2^2 \quad \text{and} \quad k^2 - a_1^2 + a_2^2$$

are both perfect squares. That is, $b^2 - a^2$ must be additively closed across four of the rectangles.

Proof. By Corollary 13 the square closes iff the forced edges $B = 3k^2 - a_1^2 - a_2^2$ and $D = k^2 - a_1^2 + a_2^2$ are perfect squares. Writing $\delta_i = k^2 - a_i^2$, one has $B = k^2 + (\delta_1 + \delta_2)$ and $D = k^2 + (\delta_1 - \delta_2)$; each is the larger cell $k^2 + \delta$ of a rectangle exactly when $\delta_1 \pm \delta_2$ is an offset, i.e. when $\mu_1 \pm \mu_2 = 2(\delta_1 \pm \delta_2)$ is realised as some $b^2 - a^2$. $\square$

remark 25 (Angular form on the Thales circle). The two triangles share the diagonal as hypotenuse, so their right-angle apexes lie on the Thales circle of that diagonal. Parametrising an apex by its angle φ (with $s = k \sin \varphi$, $d = k \cos \varphi$) gives

$$\mu = 2k^2 \sin(2\varphi),$$

so the closure rule is the angular condition

$$\sin(2\varphi_1) + \sin(2\varphi_2) = \sin(2\varphi_3), \quad \sin(2\varphi_1) - \sin(2\varphi_2) = \sin(2\varphi_4).$$

Four of the $N(k)$ apex angles must lock into this sum-and-difference relation.

corollary 14 (Sum-to-product form). Applying $\sin X \pm \sin Y = 2 \sin \frac{X \pm Y}{2} \cos \frac{X \mp Y}{2}$ to the two closure equations gives

$$2 \sin(\varphi_1 + \varphi_2) \cos(\varphi_1 - \varphi_2) = \sin 2\varphi_3, \quad 2 \cos(\varphi_1 + \varphi_2) \sin(\varphi_1 - \varphi_2) = \sin 2\varphi_4,$$

and adding, then subtracting, the original pair yields the dual relations

$$\sin 2\varphi_3 + \sin 2\varphi_4 = 2 \sin 2\varphi_1, \quad \sin 2\varphi_3 - \sin 2\varphi_4 = 2 \sin 2\varphi_2,$$

equivalently

$$\sin(\varphi_3 + \varphi_4) \cos(\varphi_3 - \varphi_4) = \sin 2\varphi_1, \quad \cos(\varphi_3 + \varphi_4) \sin(\varphi_3 - \varphi_4) = \sin 2\varphi_2.$$

In the eccentricities this is linear: $\mu_3 + \mu_4 = 2\mu_1$ and $\mu_3 - \mu_4 = 2\mu_2$, so μ_1, μ_2 are the mean and half-gap of the pair $\{\mu_3, \mu_4\}$ — the closure $\{\mu_1 + \mu_2, \mu_1 - \mu_2\} = \{\mu_3, \mu_4\}$ read in reverse.

For $k = 8840$ (thirteen rectangles) no four apexes satisfy it: for the pair 2936×12152 and 4760×11560 , both $\mu_1 + \mu_2$ and $|\mu_1 - \mu_2|$ fall between marks, matching no other rectangle. The area window and the count $N(k)$ are settled by geometry and by the primes $\equiv 1 \pmod{4}$; the additive closure of $b^2 - a^2$ is the remaining — and hardest — relation, equivalent to a rational point on the associated elliptic curve, and it is what a 3×3 magic square of nine squares would require.

28 Breaking conditions: the sine ceiling and the lattice

The closure of Theorem 16 fails in two distinct ways, both visible once μ is written through the law of sines.

Proposition 40 (Law-of-sines form). The diagonal is the circumdiameter of the Thales circle, $k\sqrt{2} = 2R$, so each leg is $2R$ times the sine of its opposite angle: $a = 2R \sin \alpha$, $b = 2R \sin \beta$ with $\alpha + \beta = \frac{\pi}{2}$. Hence

$$\mu = b^2 - a^2 = 4R^2 \sin(\alpha + \beta) \sin(\beta - \alpha) = 2k^2 \sin(\beta - \alpha),$$

i.e. $\mu = 2k^2 \sin 2\varphi$ with $2\varphi = \beta - \alpha$.

The conditions in φ

Using $\sin 2\varphi = 2 \sin \varphi \cos \varphi$ and the lattice point $(s, d) = (k \sin \varphi, k \cos \varphi)$,

$$\mu = 4k^2 \sin \varphi \cos \varphi = 4sd, \quad \delta = 2k^2 \sin \varphi \cos \varphi = 2sd,$$

and $sd/2$ is the triangle's area. The closure rule reads

$$\sin \varphi_1 \cos \varphi_1 \pm \sin \varphi_2 \cos \varphi_2 = \sin \varphi_3 \cos \varphi_3, \quad \sin \varphi_4 \cos \varphi_4 \quad \Longleftrightarrow \quad s_1 d_1 \pm s_2 d_2 = s_3 d_3, \quad s_4 d_4.$$

corollary 15 (Break 1: amplitude / sine ceiling). Since $\sin \varphi \cos \varphi \leq \frac{1}{2}$ (maximum at $\varphi = \frac{\pi}{4}$, the right-isosceles triangle $s = d = k/\sqrt{2}$), every eccentricity obeys $\mu \leq 2k^2$, equivalently $sd \leq k^2/2$. The closure is therefore impossible whenever

$$\sin \varphi_1 \cos \varphi_1 + \sin \varphi_2 \cos \varphi_2 > \frac{1}{2} \quad \Longleftrightarrow \quad s_1 d_1 + s_2 d_2 > \frac{k^2}{2} :$$

no angle φ_3 has $\sin \varphi_3 \cos \varphi_3$ above $\frac{1}{2}$, so the required apex leaves the circle and the cell $k^2 - (\delta_1 + \delta_2)$ goes negative. Equivalently, the two triangle areas cannot sum past the maximal $k^2/4$.

corollary 16 (Break 2: the lattice). Even when $sd_1 + sd_2 \leq k^2/2$, the resulting φ_3 must be an actual apex: $\sin \varphi_3 = s_3/k$, $\cos \varphi_3 = d_3/k$ with (s_3, d_3) a Gaussian-integer point on the circle. A generic sum lands strictly between apexes and realises no rectangle.

For $k = 8840$ both breaks bite in turn. The largest offset is 99.6% of k^2 , so of the $\binom{13}{2} = 78$ pairs, 61 already violate Break 1 ($\delta_1 + \delta_2 > k^2$); of the 17 survivors, every $s_1 d_1 \pm s_2 d_2$ misses the lattice (Break 2), leaving 0 magic squares. Thus a square survives only for rectangles eccentric enough to stay under the sine ceiling *and* whose leg-products land on genuine apexes — the amplitude bound is geometric (the diameter fixes $2k^2$ by the law of sines), while the lattice bound is the arithmetic, elliptic-curve obstruction.

29 Normalised form: square side 1

Setting the square side to 1 ($k = 1$) reduces every quantity to a function of the single angle φ . The base is the hypotenuse of the right triangle, and its legs are

$$P = \cos \varphi (= d), \quad Q = \sin \varphi (= s), \quad P^2 + Q^2 = 1 = k^2.$$

Proposition 41 (Sides as $\sqrt{2} \sin(45^\circ \mp \varphi)$). With $k = 1$,

$$P + Q = \cos \varphi + \sin \varphi = \sqrt{2} \sin(45^\circ + \varphi), \quad P - Q = \cos \varphi - \sin \varphi = \sqrt{2} \sin(45^\circ - \varphi),$$

and these are exactly the rectangle sides $b = P + Q$, $a = P - Q$. The restriction $\varphi < 45^\circ$ is precisely what keeps $P - Q = \sqrt{2} \sin(45^\circ - \varphi) > 0$; at $\varphi = 45^\circ$ the short side vanishes (the sliver) and at $\varphi = 0$ both sides equal 1 (the square).

The derived quantities collapse to $\sin 2\varphi$:

$$(P + Q)^2 + (P - Q)^2 = 2 \quad (a^2 + b^2 = 2), \quad 2PQ = \sin 2\varphi = \delta, \quad 4PQ = 2 \sin 2\varphi = \mu,$$

so the offset is $\delta = \sin 2\varphi$ and the eccentricity $\mu = 2 \sin 2\varphi$. The magic line centred at $k^2 = 1$ reads

$$a^2 = (P - Q)^2 = 1 - \sin 2\varphi, \quad k^2 = 1, \quad b^2 = (P + Q)^2 = 1 + \sin 2\varphi,$$

summing to $3 = 3k^2$. Thus in normalised units each apex sits at offset $\sin 2\varphi$ from the centre, its sides are $\sqrt{2} \sin(45^\circ \mp \varphi)$, and the closure and breaking conditions of the previous sections are read directly off $\sin 2\varphi$ with $0 \leq \varphi < 45^\circ$.

30 Closing summary: four unit-hypotenuse triangles

Fixing the hypotenuse to 1 collapses the whole problem onto a single angle per apex. Each apex is a right triangle with $\text{hyp} = 1$ and legs

$$P = \cos \varphi, \quad Q = \sin \varphi, \quad P^2 + Q^2 = 1,$$

whose rectangle has long side $P + Q = \cos \varphi + \sin \varphi$, short side $P - Q = \cos \varphi - \sin \varphi$, and offset $2PQ = \sin 2\varphi$, all with $0 \leq \varphi < 45^\circ$.

Proposition 42 (The problem on the unit-hypotenuse arc). A 3×3 magic square of squares centred at k^2 (normalised $k = 1$) is equivalent to four angles $\varphi_1, \varphi_2, \varphi_3, \varphi_4 \in [0, 45^\circ)$, each a right triangle of hypotenuse 1, whose offsets close additively:

$$\sin 2\varphi_3 = \sin 2\varphi_1 + \sin 2\varphi_2, \quad \sin 2\varphi_4 = \sin 2\varphi_1 - \sin 2\varphi_2,$$

with each $\cos \varphi_i \pm \sin \varphi_i$ a genuine apex (an integer side after rescaling by k). Two angles are chosen (the diagonals); the other two are forced.

The square is then read off directly:

line	cells ($1 \mp \sin 2\varphi$)
main diagonal (φ_1)	$1 - \sin 2\varphi_1, 1, 1 + \sin 2\varphi_1$
anti-diagonal (φ_2)	$1 - \sin 2\varphi_2, 1, 1 + \sin 2\varphi_2$
middle column (φ_3)	$1 - \sin 2\varphi_3, 1, 1 + \sin 2\varphi_3$
middle row (φ_4)	$1 - \sin 2\varphi_4, 1, 1 + \sin 2\varphi_4$

each line summing to $3 = 3k^2$. The two conditions are the amplitude bound $\sin 2\varphi_1 + \sin 2\varphi_2 \leq 1$ (so φ_3 exists in the arc, Corollary 15) and the lattice bound (each forced angle a Gaussian apex, Corollary 16). Thus the entire magic-square-of-squares question is: *find four right triangles of unit hypotenuse whose $\sin 2\varphi$ offsets form a closed set $\{s, t, s + t, s - t\}$ on the 45° arc* — two free, two forced — a search that, across every centre examined, meets the amplitude ceiling or the lattice, never both at once.

The relation among the four angles: areas add

The condition binding $\varphi_1, \varphi_2, \varphi_3, \varphi_4$ is not linear in the angles but additive in $\sin 2\varphi$, which reads geometrically as the *areas* of the four unit-hypotenuse triangles.

corollary 17 (Area closure). A unit-hypotenuse right triangle has area $A = \frac{1}{2} \cos \varphi \sin \varphi = \frac{1}{4} \sin 2\varphi$, so the offset is $\sin 2\varphi = 4A$ and the closure of Proposition 42 becomes

$$A_3 = A_1 + A_2, \quad A_4 = A_1 - A_2 :$$

the two forced triangles have areas equal to the sum and difference of the two chosen triangles' areas. Equivalently

$$\sin 2\varphi_1 = \frac{1}{2}(\sin 2\varphi_3 + \sin 2\varphi_4), \quad \sin 2\varphi_2 = \frac{1}{2}(\sin 2\varphi_3 - \sin 2\varphi_4).$$

remark 26 (The angles themselves do not add). The relation is on areas, not angles: $\varphi_3 \neq \varphi_1 + \varphi_2$ in general, since

$$\sin(2\varphi_1 + 2\varphi_2) = \sin 2\varphi_1 \cos 2\varphi_2 + \cos 2\varphi_1 \sin 2\varphi_2 \neq \sin 2\varphi_1 + \sin 2\varphi_2,$$

the surplus being exactly the cosine factors. In sum-to-product form the constraint is

$$\sin(\varphi_1 + \varphi_2) \cos(\varphi_1 - \varphi_2) = \frac{1}{2} \sin 2\varphi_3, \quad \cos(\varphi_1 + \varphi_2) \sin(\varphi_1 - \varphi_2) = \frac{1}{2} \sin 2\varphi_4.$$

Thus the four angles are locked by a single geometric law — pick two unit-hypotenuse triangles, and the sum and difference of their areas must again be areas of triangles in the arc (all four genuine lattice apexes). Additive on areas, transcendental on angles: this is the precise relation among $\varphi_1, \varphi_2, \varphi_3, \varphi_4$.

31 The sum is a cosine: the boundary obstruction

Adding and subtracting the two side-terms of Proposition 41 collapses them by sum-to-product:

$$f(\varphi) = \sin(45^\circ + \varphi) + \sin(45^\circ - \varphi) = 2 \sin 45^\circ \cos \varphi = \sqrt{2} \cos \varphi,$$

$$g(\varphi) = \sin(45^\circ + \varphi) - \sin(45^\circ - \varphi) = 2 \cos 45^\circ \sin \varphi = \sqrt{2} \sin \varphi,$$

so $f = \sqrt{2} P$ (a cosine) and $g = \sqrt{2} Q$ (a sine): the *sum* of the two terms is a cosine, the *difference* a sine.

remark 27 (Why closures are pushed to the forbidden edge). The curves meet, $f = g$, only where $\cos \varphi = \sin \varphi$, i.e. at $\varphi = 45^\circ$ with common value $1/\sqrt{2}$ — exactly the degenerate, forbidden boundary ($a = P - Q \rightarrow 0$). In the closure one needs a sum of offsets, which behaves like the bounded, peaking cosine f , to equal another sine-offset; the only clean meeting point is the crossing $\sin = \cos = 1/\sqrt{2}$, the angle that degenerates the rectangle and is therefore excluded. Genuine closures are thus forced toward a boundary they cannot reach — the geometric face of the amplitude ceiling $\sin 2\varphi_1 + \sin 2\varphi_2 \leq 1$.

The correct search runs the closure over the true integer apexes, not over continuous angles:

```

from math import isqrt
def search_closure(kmax):
    """A full 9-square needs four apexes whose sin2phi offsets close:
        s3 = s1 + s2 and s4 = s1 - s2 (s_i = sin2phi_i), all real apexes,
        with (P+Q) <= k^2 (amplitude) and (P +/- Q) genuine offsets (
            lattice)."""
    for k in range(2, kmax):
        c = k*k
        offs = {c-a*a for a in range(1,k)
                if isqrt(2*c-a*a)**2 == 2*c-a*a and isqrt(2*c-a*a) > k}
        if len(offs) < 4: continue
        ol = sorted(offs)
        for P in ol:
            for Q in ol:
                if Q >= P: break
                if P+Q <= c and (P+Q) in offs and (P-Q) in offs:
                    return k, (P, Q, P+Q, P-Q) # a real nine-square
    return None

```

Run to $k < 8000$ it returns `None`: no closure, no 3×3 magic square of nine squares — consistent with the boundary obstruction above and with the problem's status as open. The $f = \sqrt{2}\cos$, $g = \sqrt{2}\sin$ identity is the *explanation*; the lattice search `search_closure` is the *test*; and the forbidden crossing at $1/\sqrt{2}$ is where the two meet.

32 The condition in integers: Gaussian closure

The continuous equation $\sin x_1 + \sin x_2 = \sin x_3$ carries no constraint — for any x_1, x_2 with $\sin x_1 + \sin x_2 \leq 1$ one may take $x_3 = \arcsin(\sin x_1 + \sin x_2)$. The entire condition is *integrality*: each $x_i = 2\varphi_i$ must be the doubled angle of an integer lattice point on the circle of radius k ,

$$\sin \varphi_i = \frac{s_i}{k}, \quad \cos \varphi_i = \frac{d_i}{k}, \quad s_i, d_i \in \mathbf{Z}, \quad s_i^2 + d_i^2 = k^2.$$

Theorem 17 (Integer form of the closure). Since the offset is $\delta_i = k^2 \sin 2\varphi_i = 2s_i d_i$, a 3×3 magic square of nine squares centred at k^2 requires four lattice points (s_i, d_i) on $s^2 + d^2 = k^2$ whose leg-products close additively:

$$s_3 d_3 = s_1 d_1 + s_2 d_2, \quad s_4 d_4 = s_1 d_1 - s_2 d_2.$$

No real-angle condition survives; the constraint is entirely this Diophantine one.

For $k = 1105$ the thirteen lattice points and their products $s_i d_i$ are

s	d	sd	s	d	sd
47	1104	51888	520	975	507000
105	1100	115500	561	952	534072
169	1092	184548	576	943	543168
264	1073	283272	663	884	586092
272	1071	291312	700	855	598500
425	1020	433500	744	817	607848
468	1001	468468			

No four of these products satisfy $s_3d_3 = s_1d_1 \pm s_2d_2$: the additive quadruple has zero solutions, so 1105 admits no nine-square magic square.

remark 28 (Gaussian form). Each (s, d) is a Gaussian integer $z = s + id$ with $|z|^2 = k^2$, and

$$z^2 = (d^2 - s^2) + i(2sd), \quad \text{Im}(z^2) = 2sd = \delta,$$

so the offsets are the imaginary parts of the squares of the norm- k^2 Gaussian integers, all on the circle $|w| = k^2$. The magic-square condition is four such integers whose squares have additively closed imaginary parts,

$$\text{Im}(z_3^2) = \text{Im}(z_1^2) + \text{Im}(z_2^2), \quad \text{Im}(z_4^2) = \text{Im}(z_1^2) - \text{Im}(z_2^2).$$

This is the elliptic-curve condition in its cleanest form. The angles $\arctan(s/d)$ merely label these Gaussian integers; the real constraint is that their squared imaginary parts add. Across every centre examined no additive quadruple occurs — the whole problem, seen entirely in integers, and still open.

33 Synthesis: four inscribed triangles whose areas add

Every apex is a right triangle inscribed in the circle: legs (s, d) , hypotenuse k , and since the hypotenuse is the diameter, all such triangles share one circumcircle (Thales). Its area is a quarter of its offset,

$$A = \frac{1}{2}sd, \quad \delta = 2sd = 4A.$$

Theorem 18 (The problem as inscribed-triangle areas). A 3×3 magic square of nine squares centred at k^2 is equivalent to four right triangles with integer legs, inscribed in a single circle of hypotenuse k , whose areas close additively:

$$A_3 = A_1 + A_2, \quad A_4 = A_1 - A_2.$$

Two triangles are chosen; the sum and difference of their areas must again be areas of triangles on the same circle.

The boundary case $s = d$ is the isosceles right triangle of maximal area $k^2/4$ — the degenerate, forbidden apex ($\varphi = 45^\circ$, $\sin \varphi = \cos \varphi = 1/\sqrt{2}$, offset at the ceiling k^2). It requires $k = s\sqrt{2}$, an irrational radius, so the perfectly balancing triangle lies at a point no integer circle contains. This is the geometric reason the closure is forever pushed toward a boundary it cannot reach.

remark 29 (Congruent numbers). A congruum — the common difference of three squares in arithmetic progression — is four times the area of a right triangle. “Areas that add on one circle” is therefore a system of congrua, the classical elliptic-curve object. The journey from the $6k \pm 1$ wheel and the base-sixty inheritance, through Pythagorean triples and the $\sin 2\varphi$ offsets, terminates here: in four integer-legged right triangles inscribed in a single circle whose areas add. None has ever been found, and the reason is visible — the perfect balancing triangle ($s = d$, area $k^2/4$) sits at the irrational radius $k = s\sqrt{2}$, just outside the reach of the lattice.

34 The maximal apex: half the inscribed square

The inscribed right triangles are bounded in area, and the bound has a clean shape.

Proposition 43 (Maximal inscribed right triangle). Among right triangles with hypotenuse equal to the diameter k , the area is maximal for the isosceles one $s = d = k/\sqrt{2}$, with

$$A_{\max} = \frac{k^2}{4} \quad \Longleftrightarrow \quad \delta_{\max} = 4A_{\max} = k^2 \quad (\sin 2\varphi = 1),$$

so the offset ceiling k^2 is exactly four times the maximal area.

Geometrically this maximal triangle is *half the square inscribed in the circle*: the inscribed square has its diagonal on the diameter k , hence side $k/\sqrt{2}$ and area $k^2/2$; cutting it along a diagonal yields two isosceles right triangles of area $k^2/4$ each. Every genuine apex triangle is a scalene ($s \neq d$) “leaning” version of this half-square, with area just below $k^2/4$ — for $k = 1105$ the largest is (744, 817) with area $303924 < k^2/4 = 305256.25$. The perfect half-square is never reached because it needs $s = d$, i.e. $k = s\sqrt{2}$, an irrational radius: the maximal, balancing apex sits at the same forbidden point ($\varphi = 45^\circ$, $\sin \varphi = \cos \varphi = 1/\sqrt{2}$) that the closure is forever pushed toward.

35 A proof target

Theorem 18 reduces the existence of a 3×3 magic square of nine squares to a purely Diophantine statement about inscribed triangles. This suggests a concrete target for a proof of non-existence.

Proof target. Show that the system

$$s_3d_3 = s_1d_1 + s_2d_2, \quad s_4d_4 = s_1d_1 - s_2d_2, \quad s_i^2 + d_i^2 = k^2 \quad (i = 1, 2, 3, 4),$$

has no solution in positive integers for any k (with the induced nine cells distinct).

Equivalently: no four right triangles with integer legs, inscribed in a common circle,

have additively closed areas $A_3 = A_1 + A_2$, $A_4 = A_1 - A_2$.

Proposition 44 (Numerical evidence). Not a single circle $s^2 + d^2 = k^2$ with $k < 7000$ admits even a *partial* area-relation: there is no triple of inscribed integer right triangles with $A_i \pm A_j = A_k$. In particular the full quadruple never occurs, so no nine-square magic square exists with $k < 7000$; and the `search_closure` scan confirms no closure through $k < 8000$.

The strength of this evidence is worth stating plainly: it is not that the four-triangle closure *narrowly* fails, but that inscribed integer right triangles on one circle appear never to have areas that add *even two at a time*. A proof of non-existence would upgrade this observed additive independence to a theorem.

remark 30 (Where a proof would live). The single relation $s_1d_1 = s_2d_2 + s_3d_3$ with all points on $s^2 + d^2 = k^2$ is a statement about *concordant forms* (Euler) and rational points on an associated elliptic curve; the products $s_id_i = \frac{1}{2} \operatorname{Im}(z_i^2)$ are imaginary parts of squared Gaussian integers of norm k^2 (Remark following Theorem 17). A proof of the target would show that these imaginary parts cannot be additively closed in fours — a bound on the rank or torsion of the relevant curve — and would resolve the classical open problem. Conversely a single integer solution would produce the first known 3×3 magic square of nine squares. The reduction is complete; only this last, number-theoretic step remains open.

36 A dictionary of the apex quantities

Two right triangles meet at each apex, and the area formula uses the legs of the inscribed one — not the cell roots. Confusing the two is easy, so we record the conversions explicitly.

symbol	meaning
k	hypotenuse = circle diameter; centre cell = k^2
a, b	cell roots: low cell = a^2 , high cell = b^2 , $a^2 + b^2 = 2k^2$
$s = \frac{b-a}{2}$	leg of the inscribed triangle (half- <i>difference</i> of roots)
$d = \frac{b+a}{2}$	leg of the inscribed triangle (half- <i>sum</i> of roots), $s^2 + d^2 = k^2$
$\delta = 2sd$	offset = $k^2 - a^2 = b^2 - k^2$
$A = \frac{1}{2}sd$	area of the inscribed triangle = $\delta/4$

The legs s, d are *midpoint* quantities of the roots, not the roots themselves; one must convert $a, b \mapsto s = \frac{b-a}{2}$, $d = \frac{b+a}{2}$ before applying $A = \frac{1}{2}sd$. Because

$$sd = \frac{(b-a)(b+a)}{4} = \frac{b^2 - a^2}{4},$$

the product sd (and hence the area) is an integer whenever the two cells are integers — *even when s, d themselves are irrational*.

Example 9 (The half-lattice apex 373/222121 of $k = 425$). Here $a = 373$ (so $a^2 = 139129 = 373^2$, a square) and $b = \sqrt{222121} = 471.297\dots$ (irrational, so $b^2 = 222121$ is not a square). Then

$$s = \frac{b-a}{2} = 49.148\dots, \quad d = \frac{b+a}{2} = 422.148\dots \text{ (both irrational),}$$

yet

$$sd = \frac{222121 - 139129}{4} = 20748 \text{ (integer),} \quad A = \frac{1}{2}sd = 10374 \text{ (integer),} \quad a^2 + b^2 = 361250 = 2 \cdot 425^2.$$

The area is a clean integer and one root, $a = 373$, is an integer; only the second root b and the legs s, d are irrational. A genuine full apex requires s and d *both* integer, so that $a = d - s$ and $b = d + s$ are both whole and both cells are squares. Rounding s, d before multiplying (e.g. $49.15 \cdot 422.15$) produces the spurious 20748.67; the exact product is $(b^2 - a^2)/4 = 20748$.

37 Constructing the apex from (k, δ)

The triangle need not be built from the cell-squares; the circle k and the offset δ determine it completely.

Proposition 45 (Construction from the offset). On the circle $s^2 + d^2 = k^2$, an apex of offset $\delta = 2sd$ has

$$\text{area } A = \frac{\delta}{4}, \quad \text{one leg from the other } d = \frac{\delta}{2s}, \quad s = \frac{\delta}{2d},$$

and the two legs are fixed by

$$s + d = \sqrt{k^2 + \delta}, \quad d - s = \sqrt{k^2 - \delta},$$

so the cell roots are $a = d - s = \sqrt{k^2 - \delta}$ and $b = d + s = \sqrt{k^2 + \delta}$, i.e. the two cells are simply $k^2 \mp \delta$.

Proof. $A = \frac{1}{2}sd = \frac{1}{4}(2sd) = \delta/4$, and $d = \delta/(2s)$ is $\delta = 2sd$ rearranged. From $sd = \delta/2$ and $s^2 + d^2 = k^2$, $(s + d)^2 = k^2 + \delta$ and $(d - s)^2 = k^2 - \delta$; taking roots gives $b = d + s$ and $a = d - s$, whence $a^2 = k^2 - \delta$, $b^2 = k^2 + \delta$. $\square$

Thus the recipe is: *choose an offset* $\delta \leq k^2$; the area is $\delta/4$, the legs have product $\delta/2$ (one obtained from the other by $\delta/(2\text{leg})$), and the two cells fall out as $k^2 \mp \delta$ — one does not insert the squares, they emerge. For the 373/222121 apex of $k = 425$, $\delta = 41496$ gives $A = 41496/4 = 10374$, $d = 41496/(2s) = 422.15\dots$, and cells $180625 \mp 41496 = 139129, 222121$. The magic-square question then reduces to the single arithmetic test on the chosen offsets: are $k^2 \mp \delta$ perfect squares — equivalently, are the legs s, d integers.

38 The three-parameter form (k, γ_1, γ_2)

By Proposition 45 an apex is fixed by two parameters — the centre k and the shift $\gamma = \delta$ (offset):

$$\text{apex}(k, \gamma) : \quad A = \frac{\gamma}{4}, \quad sd = \frac{\gamma}{2}, \quad s^2 + d^2 = k^2, \quad \text{cells } k^2 \mp \gamma.$$

Theorem 19 (The magic square as three integers). A 3×3 magic square of nine squares is determined by three integers (k, γ_1, γ_2) : one centre k and two chosen shifts γ_1, γ_2 , the remaining shifts being forced,

$$\gamma_3 = \gamma_1 + \gamma_2 \quad (\text{middle column}), \quad \gamma_4 = \gamma_1 - \gamma_2 \quad (\text{middle row}).$$

It is a square of nine squares precisely when

$$\gamma_1 + \gamma_2 \leq k^2 \quad (\text{amplitude}), \quad k^2 \mp \gamma_i \text{ are perfect squares for all } \gamma_i \in \{\gamma_1, \gamma_2, \gamma_1 + \gamma_2, \gamma_1 - \gamma_2\}.$$

The search space collapses to *three integers*, and the entire question is whether the eight values $k^2 \mp \gamma_i$ are simultaneously perfect squares. Equivalently, by the dictionary, whether the four apex leg-pairs s_i, d_i are all integers on the single circle $s^2 + d^2 = k^2$ — the Gaussian / elliptic-curve condition of Theorem 17. Fixing the centre and two shifts builds the rest of the square automatically; only squareness remains, and that residue is the open problem, now in its most compact form:

$$\text{find } (k, \gamma_1, \gamma_2) \in \mathbf{Z}^3 \text{ with all of } k^2 \mp \gamma_1, k^2 \mp \gamma_2, k^2 \mp (\gamma_1 + \gamma_2), k^2 \mp (\gamma_1 - \gamma_2) \text{ square.}$$

39 The closure is definitional; only additions remain

In the three-parameter form the area closure is not something to prove — it is built into the parametrization.

Proposition 46 (Closure is definitional). With $A_i = \gamma_i/4$ and the *definition* $\gamma_3 := \gamma_1 + \gamma_2$,

$$A_3 = \frac{\gamma_3}{4} = \frac{\gamma_1 + \gamma_2}{4} = A_1 + A_2$$

holds identically. There is nothing to prove in $A_3 = A_1 + A_2$; it is the meaning of γ_3 .

remark 31 (No subtraction — only addition). The remaining relation $\gamma_4 = \gamma_1 - \gamma_2$ is likewise an addition read backwards, $\gamma_1 = \gamma_4 + \gamma_2$. Hence the three shifts

$$\gamma_4 = \gamma_1 - \gamma_2, \quad \gamma_1, \quad \gamma_3 = \gamma_1 + \gamma_2$$

form an arithmetic progression with common difference γ_2 ; equivalently the areas A_4, A_1, A_3 are in arithmetic progression with common difference A_2 . The four shifts are the progression $\{\gamma_1 - \gamma_2, \gamma_1, \gamma_1 + \gamma_2\}$ together with its step γ_2 — for $k = 425$, the AP 97104, 138600, 180096 of step 41496 = γ_2 . No subtraction is required; the whole closure is two additions.

Theorem 20 (The genuine target). Since the closure is automatic, a 3×3 magic square of nine squares exists if and only if there are integers (k, γ_1, γ_2) with $\gamma_1 + \gamma_2 \leq k^2$ for which all eight numbers

$$k^2 \mp \gamma_1, \quad k^2 \mp \gamma_2, \quad k^2 \mp (\gamma_1 + \gamma_2), \quad k^2 \mp (\gamma_1 - \gamma_2)$$

are perfect squares. The entire difficulty is this simultaneous squareness; the area relations $A_3 = A_1 + A_2$, $A_4 = A_1 - A_2$ contribute nothing to it.

For $(k, \gamma_1, \gamma_2) = (425, 138600, 41496)$ six of the eight are squares ($205^2, 565^2, 373^2, 23^2, 289^2, 527^2$) and two are not ($222121, 360721$): the seven-square record. A proof of non-existence must show these eight can never be simultaneously square; a single witness would be the first nine-square magic square. Everything else in the construction is free.

40 A parity constraint: every cell $\equiv 1 \pmod{8}$

The mod-8 structure pins the whole square once one reduces to the primitive case.

Theorem 21 (Odd centre forces all cells $\equiv 1 \pmod{8}$). If k is odd, every cell of the square is an odd square $\equiv 1 \pmod{8}$.

Proof. Squares are $\equiv 0, 1, 4 \pmod{8}$. For odd k , $k^2 \equiv 1$, so each opposite pair satisfies $a^2 + b^2 = 2k^2 \equiv 2 \pmod{8}$. Among $\{0, 1, 4\}$ the only sum $\equiv 2$ is $1 + 1$; hence $a^2 \equiv b^2 \equiv 1$ and a, b are odd. This holds for all four pairs, and $k^2 \equiv 1$ for the centre, so all nine cells are $\equiv 1 \pmod{8}$. $\square$

corollary 18 (Reduction to odd k). If k is even then $2k^2 \equiv 0 \pmod{8}$, forcing every pair to be $\{0, 0\}$ or $\{4, 4\} \pmod{8}$ — all cells even, hence divisible by 4; dividing the square by 4 gives a smaller magic square of squares. Thus every magic square of squares reduces to a primitive one with k odd, in which (by Theorem 21) all nine cells are $\equiv 1 \pmod{8}$.

For $k = 425$ all nine cells are $\equiv 1 \pmod{8}$ (42025, 319225, 139129, 222121, 529, 360721, 83521, 277729, 180625), including the two non-square cells. Combined with the mod-10 fact that odd squares end in 1, 5, or 9, the primitive square is heavily constrained: only odd centres need be searched, and any shift γ with $k^2 \mp \gamma \not\equiv 1 \pmod{8}$ is rejected at once. This prunes the search sharply, though it does not decide existence.

41 Three views, two knobs, six conditions

The construction has appeared in three coordinate systems: the shifts (γ_1, γ_2) of Theorem 19, the two free cells (A, D) of Corollary 13, and — after normalising the centre — the scale-free ratios (α, δ) . This section records that all three are the *same* two-parameter family, that the magic (sum) conditions cost no freedom, and that the two shift-dependencies exhaust the freedom before the remaining cells can be aimed at squares. The upshot is a clean count: two knobs against six squareness conditions.

41.1 One two-parameter family in three coordinates

Proposition 47 (The three views coincide). Fix the centre $E = k^2$. Each of the following determines the same eight border cells, and the three are related by invertible changes of variable:

$$\text{shifts } (\gamma_1, \gamma_2), \quad \text{cells } (A, D), \quad \text{ratios } (\alpha, \delta) = \left(\frac{A}{k^2}, \frac{D}{k^2} \right).$$

Indeed a cell is $k^2 \mp \gamma_i$ for one of the four shifts, so $A = k^2 - \gamma_1$ and $D = k^2 - \gamma_4$ recover (γ_1, γ_2) from (A, D) ; and dividing every cell by k^2 sends the centre to 1 and (A, D) to (α, δ) .

In every one of these coordinates the family is *two*-dimensional: whichever pair one turns, the other two shifts $\gamma_3 = \gamma_1 + \gamma_2$ and $\gamma_4 = \gamma_1 - \gamma_2$ are forced (Theorem 19).

41.2 The sum is free and the centre is only a scale

Proposition 48 (The mean condition is identically satisfied). For any (A, D) the eight border cells sum to $8k^2$; equivalently the centre is their mean,

$$\frac{1}{8}(A^2 + B^2 + C^2 + D^2 + F^2 + G^2 + H^2 + I^2) = k^2, \quad \text{i.e.} \quad \sum_{\text{border}} \frac{\text{cell}}{8k^2} = 1.$$

This holds identically in (k, A, D) and imposes no constraint: the opposite cells pair to $2k^2$, and four such pairs give $8k^2$.

Proposition 49 (Scale-invariance). If a centre k^2 carries a magic square of squares, so does m^2k^2 for every m , by multiplying all nine cells by m^2 (a square times a square is a square). Hence k is a pure scale: normalising the centre to 1 removes it, leaving the two ratios (α, δ) as the only genuine parameters.

Propositions 48–49 together say that neither the magic sum nor the centre buys or spends any freedom: the sum is automatic and the centre is gauge. Every degree of freedom lives in the two ratios.

41.3 Two chosen cells, six dependent cells

Theorem 22 (Two knobs determine six cells; six squareness conditions remain). Choose the two free cells A, D (both squares). Then the remaining six border cells are integer-linear forms in (k^2, A, D) ,

$$\underbrace{I = 2k^2 - A, \quad F = 2k^2 - D}_{\text{companions of the knobs}}, \quad \underbrace{C = A + D - k^2, \quad G = 3k^2 - A - D, \quad B = 4k^2 - 2A - D, \quad H = 2A + D - 2k^2}_{\text{forced by the line sums}}$$

and all eight line-sums equal $3k^2$ automatically. A magic square of nine squares exists at this centre *iff* these six dependent cells are simultaneously perfect squares — six conditions on the two knobs.

Proof. The companions follow from the opposite-pair relations $A + I = D + F = 2k^2$. Writing the top row $= 3k^2$ gives $B = 3k^2 - A - ?$; using $C = A + D - k^2$ from column three (with I, F known) and then $B = 4k^2 - 2A - D$, $H = 2k^2 - B$, $G = 2k^2 - C$ yields the stated forms. Every line-sum is then $3k^2$ by Proposition 39, so the only surviving requirement is that I, F, C, G, B, H each be square. $\square$

remark 32 (Why the dependency is the obstruction). The two forced shifts $\gamma_3 = \gamma_1 + \gamma_2$ and $\gamma_4 = \gamma_1 - \gamma_2$ are not new freedoms: they are fixed the instant (γ_1, γ_2) are chosen. Thus the two knobs are spent placing the companions I, F , after which the four cells C, G, B, H are already determined; one cannot steer them onto squares. The dependency consumes the freedom *before* the remaining cells are filled. This is the same wall as Theorem 20: two rational parameters against a system whose simultaneous-square solutions are the rational points of an elliptic curve, whose rank — not any count of knobs — decides existence.

remark 33 (Reconciliation with Corollary 13). Corollary 13 reports *two* forced cells, not six, because it holds the two diagonals fixed as *both-square* pairs — already spending four of the six conditions (A, I, C, G square) by selecting from the $N(k)$ complementary pairs. Counting instead from two free cells (A, D) makes those four visible again, giving the raw tally of six.

For $k = 425$ with $(A, D) = (205^2, 527^2)$, four of the six dependent cells land on squares,

$$I = 565^2, \quad F = 289^2, \quad C = 373^2, \quad H = 23^2,$$

and two do not, $G = 222121$ and $B = 360721$: the seven-square record. The two misses are exactly the cells that receive no freedom.

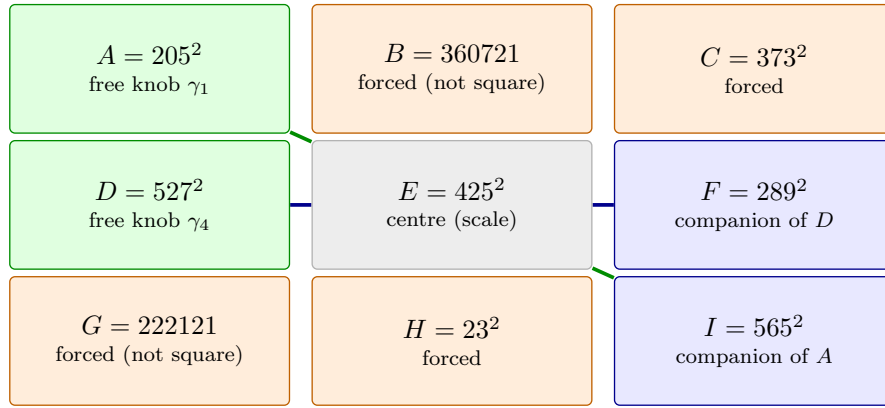


Figure 2: The two knobs and the six dependent cells at $k = 425$. The two green cells A, D are the free choices (a 2-parameter family in any of the three coordinate systems); the centre $E = k^2$ is a scale. Choosing A, D forces the two blue companions $I = 2k^2 - A$, $F = 2k^2 - D$ and then the four orange cells $C = A + D - k^2$, $G = 3k^2 - A - D$, $B = 4k^2 - 2A - D$, $H = 2A + D - 2k^2$. All eight line-sums are $3k^2$ automatically; the six coloured non-centre cells each carry an independent squareness condition. Here four are met (I, F, C, H) and two are not (G, B) — the dependent cells that receive no remaining freedom.

42 The elliptic curve: forced cells as a group law

Theorem 18 recast the problem as four right triangles on one circle $s^2 + d^2 = k^2$ whose areas close, $A_3 = A_1 + A_2$, $A_4 = A_1 - A_2$, and Theorem 17 as the additive closure of the leg-products $\delta_i = 2s_i d_i$. Additive closure of areas of triangles inscribed in a fixed circle is precisely the arithmetic of an elliptic curve — the classical theory of congrua. This section makes the dictionary explicit and exhibits the forced cells of Section 41 as *group-law images*: the two knobs are two points on the curve, and the forced cells are the points the chord-and-tangent law delivers from them.

42.1 Spokes are points; the two knobs are two points

Definition 11 (Square-point). Let E be the elliptic curve associated (Remark after Theorem 18) to area-closure on the circle $s^2 + d^2 = k^2$. A *square-point* is a rational point of E corresponding to a spoke of the square: a three-term arithmetic progression a^2, k^2, b^2 through the centre, equivalently a lattice point (s, d) with $a = d - s$, $b = d + s$ and offset $\delta = 2sd$.

Proposition 50 (The additive closure is the group law). Under the dictionary of Definition 11, the two free spokes γ_1, γ_2 — the two knobs of Theorem 22 — are two rational points $P, Q \in E$, and the two forced spokes are their group-law combinations,

$$\gamma_3 = \gamma_1 + \gamma_2 \longleftrightarrow P \oplus Q, \quad \gamma_4 = \gamma_1 - \gamma_2 \longleftrightarrow P \ominus Q.$$

Thus the shift-dependency of Section 41 is addition on E : once P and Q are chosen the forced points carry no freedom, exactly as the forced cells C, G, B, H carry none.

42.2 The forced cells are chord-and-reflect images

On a Weierstrass model $y^2 = x^3 + ax + b$ the group law is geometric: the line through P and Q meets the cubic in exactly one further point (a line meets a cubic in three), and reflecting that point across the x -axis gives $P \oplus Q$. Because the curve is degree three, this third intersection always exists and is again rational — the mechanism a conic (a parabola, degree two, two intersections) does not possess. The forced spoke, hence the forced cells, is this manufactured point: not chosen, but produced by P and Q .

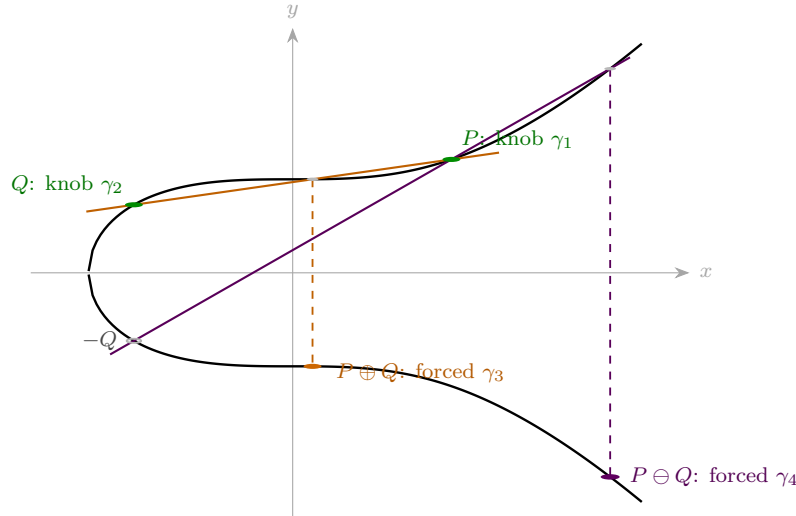


Figure 3: Both forced cells as group-law images on the associated elliptic curve (illustrative model $y^2 = x^3 + 17$). The two green points P, Q are the free spokes — the knobs γ_1, γ_2 . The orange line through P and Q meets the cubic at a third point; reflecting it gives $P \oplus Q$, the forced spoke $\gamma_3 = \gamma_1 + \gamma_2$ (cells B, H). The violet line through P and $-Q$ (the reflection of Q) gives $P \ominus Q$, the forced spoke $\gamma_4 = \gamma_1 - \gamma_2$ (cells C, G). Both forced spokes are manufactured by the group law from the two knobs; neither carries any freedom. A nine-square magic square needs both $P \oplus Q$ and $P \ominus Q$ to be square-points at once.

remark 34 (Sum and difference are a matched pair). The two forced points are not independent. Their x -coordinates are the two roots of a single quadratic symmetric in P, Q (the sum/difference addition formulas), and their canonical heights obey the parallelogram law

$$\hat{h}(P \oplus Q) + \hat{h}(P \ominus Q) = 2\hat{h}(P) + 2\hat{h}(Q).$$

So $P \oplus Q$ and $P \ominus Q$ always arrive matched — making one small forces the other large. This mirrors the shift arithmetic $\gamma_3 + \gamma_4 = 2\gamma_1$, $\gamma_3 - \gamma_4 = 2\gamma_2$: sum and difference balanced around the knobs, exactly as each cell pair balances $2k^2$ around the centre.

Where the manufactured point lands is set not by the operation but by the two knobs. Figure 4 isolates the subtraction $P \ominus Q = P \oplus (-Q)$ for a choice with Q on the lower branch: the chord through P and $-Q$ is nearly flat, so $P \ominus Q$ falls *between* the two square-points, in contrast to Figure 3, where the same operation threw the forced point far outside.

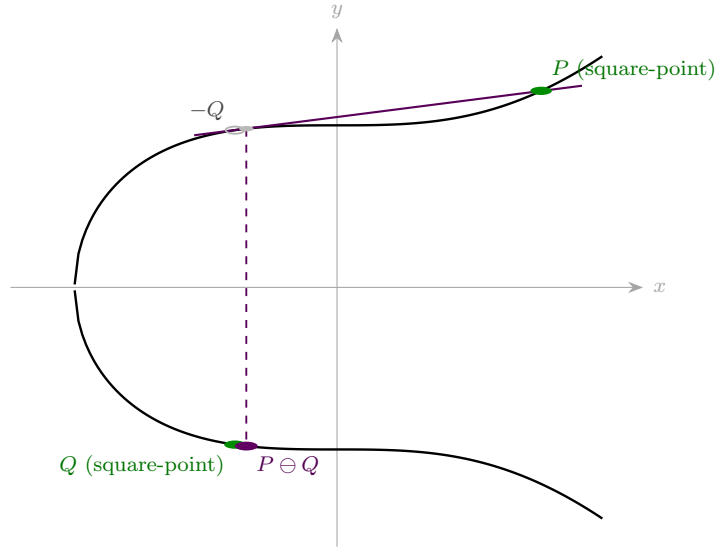


Figure 4: The subtraction view, $P \ominus Q = P \oplus (-Q)$. The second square-point Q is taken on the lower branch, so its reflection $-Q$ is nearly level with P and the chord $P - (-Q)$ is almost flat; its third intersection sits near the axis and reflecting gives $P \ominus Q = (-\frac{8}{9}, -\frac{109}{27})$, lying *between* the two square-points P and Q . The operation is the same group law as in Figure 3; only the height budget of Remark 34 decides whether the difference lands near the knobs or far beyond them.

remark 35 (Negating a knob reflects the outputs). Replacing P by $-P$ gives

$$(-P) \oplus Q = -(P \ominus Q), \quad (-P) \ominus Q = -(P \oplus Q) :$$

both outputs reflect across the x -axis and the sum swaps with the difference. On the square this is only the freedom to read each spoke from either end, $k^2 \mp \gamma$; negating a knob is the same square in mirrored labelling.

Figure 5 shows both operations from the reflected knob $-P$ at once: the two chords through $-P$ produce $(-P) \oplus Q$ and $(-P) \ominus Q$, the mirror images of the sum and difference of Figure 3.

remark 36 (Sign = branch = below/above). Because $P \ominus Q = P \oplus (-Q)$, there is only one operation; the free choice is the *sign* of each point. On the curve P and $-P$ are the two branches

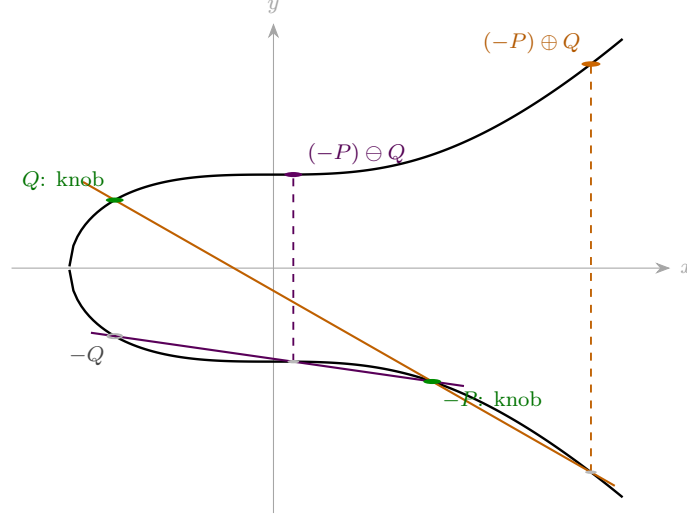


Figure 5: The reflected-knob view. Using $-P$ (the reflection of P) with the same Q , the orange chord $(-P) - Q$ gives $(-P) \oplus Q = (4, 9)$ and the violet chord $(-P) - (-Q)$ gives $(-P) \ominus Q = (\frac{1}{4}, \frac{33}{8})$. By Remark 35 these are $-(P \ominus Q)$ and $-(P \oplus Q)$: exactly the sum and difference of Figure 3 reflected across the axis, with their roles swapped. The same magic square, read from the mirrored ends of its spokes.

(same x , opposite y); on the square a spoke's two cells are $k^2 \mp \gamma$ (same gap, one below and one above the centre). These are the same choice: the sign of a point is the below/above side of its cell. A *square-point* — a cell that is a genuine perfect square — may sit on the below branch, the above branch, or both, so searching only one sign (only below roots, or only above roots) can miss a square living on the other. At $k = 425$ every spoke carries its square below and two also above:

γ	$k^2 - \gamma$	$k^2 + \gamma$	square on
138600	205^2	565^2	both
97104	289^2	527^2	both
41496	373^2	222121	below only
180096	23^2	360721	below only

so a below-only search sees all four square-points while an above-only search sees only the two both-square spokes; both nonetheless describe the same square. In general, though, a required square-point can exist on one branch alone, so both signs must be searched: the branch of the point, not the choice of $\oplus$ or $\ominus$, is what locates the square. Geometrically this is the curve's reflection symmetry in the x -axis, $y \mapsto -y$: the graph has two legs meeting at the real root, the two legs are the two signs (below and above the centre), and a square-point may lie on either leg — so one must search both legs of the curve, never only one.

remark 37 (The centre is the fixed point of the reflection). The same reflection also pins the centre. The map $(x, y) \mapsto (x, -y)$ — negation $P \mapsto -P$ — has fixed locus $y = 0$, the axis of symmetry, where the offset $\delta = 2sd$ vanishes. Under the dictionary $\delta = 0$ is the degenerate spoke $\gamma = 0$: the centre cell k^2 itself, the one cell equal to its own companion $k^2 \mp 0$. Equivalently, in the normalised value view (Proposition 49) every cell is $1 \pm \gamma/k^2$ and the antipodal pairing becomes the reflection $x \mapsto 2 - x$, whose unique fixed point is the centre 1. Hence k^2 is not a free parameter but the *fixed point* of the reflection — the mean of the eight border cells (Proposition 48), normalised

to 1 and then scaled away entirely. The two legs of the curve carry the $\pm$ spokes; the axis between them carries the centre, pinned rather than chosen.

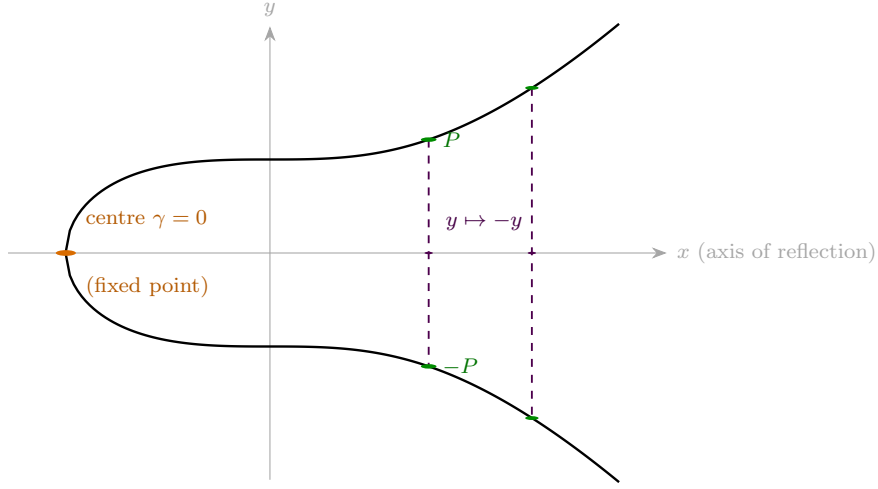


Figure 6: Why k is fixed, not chosen. The curve is symmetric across the x -axis, $(x, y) \mapsto (x, -y)$ (negation $P \mapsto -P$): every point pairs with its mirror on the other leg (green pairs, dashed connectors), so the two legs carry the $\pm$ spokes $k^2 \mp \gamma$. The single point fixed by the reflection is where the curve meets the axis, $y = 0$ — offset $\delta = 0$, the degenerate spoke $\gamma = 0$, i.e. the centre k^2 . It lies on the axis of symmetry, pinned rather than chosen; in the normalised value view it is the fixed point 1 of $x \mapsto 2 - x$ (Remark 37). So k is the reflection's fixed point, not a free parameter.

remark 38 (Locating a chord's crossing by symmetry). Where a chord meets the mirror axis is fixed by the two heights alone. With the axis as $y = 0$, dropping perpendiculars from the endpoints $A = (x_A, y_A)$, $B = (x_B, y_B)$ makes two similar right triangles sharing the crossing x_0 (equal vertical angles), so

$$\frac{x_0 - x_A}{x_B - x_0} = \frac{|y_A|}{|y_B|}, \quad x_0 = \frac{x_A y_B - x_B y_A}{y_B - y_A}.$$

For $A = -P = (2, -5)$, $B = Q = (-2, 3)$ this gives $x_0 = \frac{6-10}{8} = -\frac{1}{2}$, the run splitting $1.5 : 2.5 = 3 : 5 = |y_A| : |y_B|$. Through the Gaussian dictionary the heights $|y|$ are the offsets δ of the two spokes, so the crossing is a similar-triangles average of the two amplitudes — fixed entirely by the reflection symmetry, though (unlike the curve's own axis-crossing, the centre k^2) it is not itself a cell.

remark 39 (Crossing versus shift: a lever law). Write each endpoint as a shift from the centre axis: foot (horizontal position) u_i and *signed* shift $\sigma_i = \pm 2s_i d_i$ (positive above the centre, negative below). The crossing of Remark 38 is then a lever balanced by the two shifts,

$$x_0 = \frac{u_1 \sigma_2 - u_2 \sigma_1}{\sigma_2 - \sigma_1}, \quad \text{equivalently} \quad \sigma_2 (x_0 - u_1) = \sigma_1 (x_0 - u_2).$$

Its position is decided by the *signs* of the shifts, not their sizes: opposite signs (one spoke above the centre, one below) place x_0 *between* the feet; equal signs place it *outside*. This is once more the below/above rule of Remark 36 — the branch of each spoke governs the crossing — and the centre is the degenerate case $\sigma = 0$, the fixed axis point k^2 where the lever has no moment.

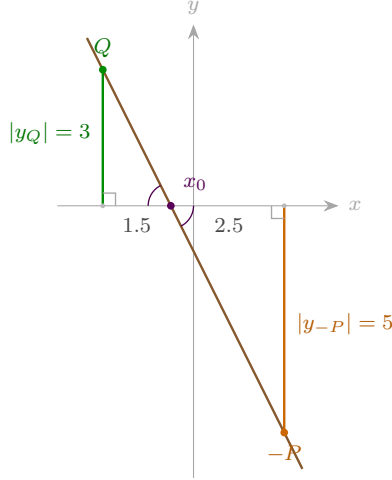


Figure 7: Locating the crossing by symmetry. The mirror axis is $y = 0$; the heights of the two endpoints, $|y_Q| = 3$ and $|y_{-P}| = 5$ (perpendiculars, right angles at their feet), form two similar right triangles sharing the crossing x_0 (equal vertical angles, marked). The run therefore splits in the ratio of the heights, $1.5 : 2.5 = 3 : 5$, giving $x_0 = (x_A y_B - x_B y_A) / (y_B - y_A) = -\frac{1}{2}$ (Remark 38). The symmetry alone — the two amplitudes measured from the axis — fixes the crossing by trigonometry.

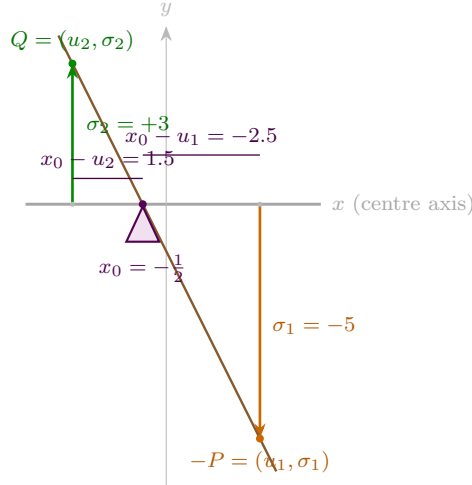


Figure 8: The lever law of Remark 39. The centre axis is the beam; the signed shifts $\sigma_2 = +3$ (above, at foot $u_2 = -2$) and $\sigma_1 = -5$ (below, at $u_1 = 2$) act as opposing moments, and the chord through the two tips crosses the axis at the balance point $x_0 = -\frac{1}{2}$. There $\sigma_2(x_0 - u_1) = \sigma_1(x_0 - u_2)$, i.e. $3(-2.5) = (-5)(1.5) = -7.5$. Opposite signs put the fulcrum *between* the feet; the centre k^2 is the zero-moment point $\sigma = 0$.

remark 40 (Two axis-anchors replace the second knob). Two points $\{P, Q\}$ determine the curve $y^2 = x^3 + ax + b$: subtracting the two membership equations gives

$$a = \frac{(y_P^2 - y_Q^2) - (x_P^3 - x_Q^3)}{x_P - x_Q}, \quad b = y_Q^2 - x_Q^3 - a x_Q.$$

But the second knob can be traded for two points on the mirror axis, $\{Q; x_0, x_1\}$, with no loss. The root x_1 (the centre) fixes the *curve*: it and Q give two *linear* equations in (a, b) ,

$$a x_1 + b = -x_1^3, \quad a x_Q + b = y_Q^2 - x_Q^3,$$

returning $(a, b) = (0, 17)$ for $Q = (-2, 3)$, $x_1 = -\sqrt[3]{17}$. The crossing x_0 then fixes the *chord*: the line through Q meeting the axis at x_0 has slope $y_Q/(x_Q - x_0)$ and re-meets the curve at P . Hence

$$\{P, Q\} \cong \{Q; x_0, x_1\}, \quad x_1 \rightsquigarrow \text{curve } (a, b), \quad x_0 \rightsquigarrow \text{chord (hence } P).$$

The two free degrees of freedom may be carried as two points, or as one point and its two axis-anchors — the centre x_1 and the crossing x_0 .

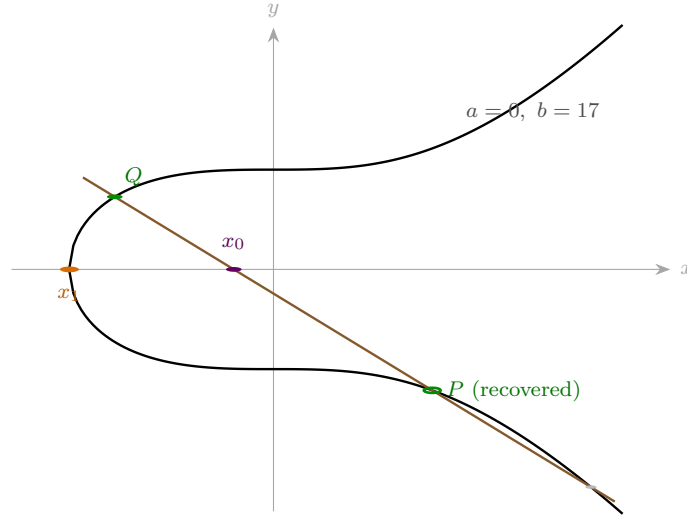


Figure 9: Two axis-anchors replace the second knob (Remark 40). From the one knob $Q = (-2, 3)$ and the root $x_1 = -\sqrt[3]{17}$ (the curve's axis-crossing, the centre), two linear equations return the curve $a = 0$, $b = 17$. The crossing $x_0 = -\frac{1}{2}$ then fixes the chord through Q (slope $y_Q/(x_Q - x_0) = -2$), which re-meets the curve at the recovered point $P = (2, -5)$ (and a third point $(4, -9)$). So $\{Q; x_0, x_1\}$ carries the same information as $\{P, Q\}$: x_1 the curve, x_0 the chord.

42.3 A circular relation: the curve buys no freedom

The axis-anchor picture must be read with care once the centre is fixed. By Theorem 19 a magic square of squares is fixed by (k, γ_1, γ_2) , of which k is a scale (Proposition 49); the essential freedom is the two spokes (γ_1, γ_2) . On the host curve the relevant quantities are three points on the mirror axis: the centre x_1 (the root, from k), and two that encode the spokes — the chord crossing x_0 and the abscissa $x_2 = x_P$ of the recovered point. They are *not* independent axes: once the curve is fixed, any two of the spoke-quantities determine the third, so they close in a loop rather than spread freely.

Proposition 51 (The spoke quantities satisfy one incidence relation). Fix the curve (equivalently the centre $x_1 = \text{root of } x^3 + ax + b$). For collinear curve points P, Q with third intersection R , the crossing x_0 and the abscissae $x_2 = x_P, x_Q$ obey the single relation

$$x_0^2 = \frac{x_P x_Q x_R + b}{x_P + x_Q + x_R}, \quad x_R = \left(\frac{y_Q - y_P}{x_Q - x_P} \right)^2 - x_P - x_Q.$$

Hence $\{x_Q, x_0, x_2\}$ carry one constraint: choosing any two fixes the third. With x_1 the fixed centre and k a scale (Proposition 49), the square keeps only the two spoke freedoms (γ_1, γ_2) , linked in a closed loop rather than on independent axes.

So x_2 is fixed once x_0 , the knob Q , and the curve x_1 are fixed (for our data $x_0 = -\frac{1}{2}$, $b = 17$ give $x_0^2 = \frac{(-16)+17}{4} = \frac{1}{4}$, returning $x_2 = 2$). The elliptic curve, degree three in x , supplies the incidence — a line meets it in three points — but grants *no* new degree of freedom: degree three is what closes the loop, not what opens it. So passing to the elliptic curve buys nothing — the square is still *two* degrees of freedom, the two spokes (γ_1, γ_2) , on the curve exactly as off it, closed in the circular relation of Figure 10.

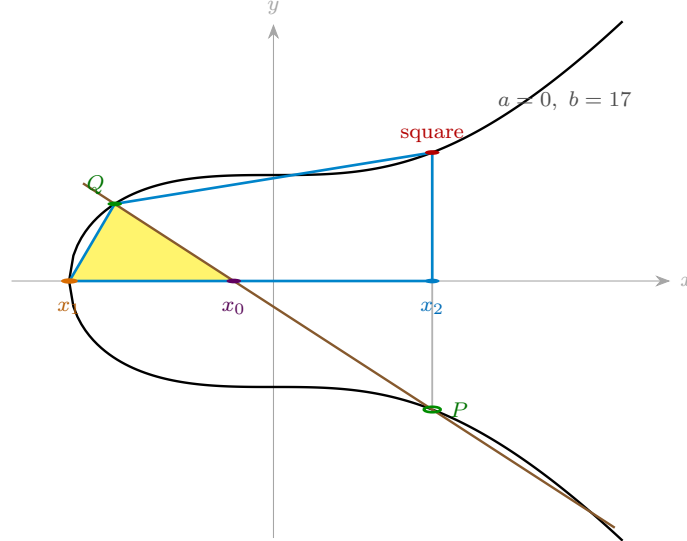


Figure 10: The circular relation (Proposition 51). The three axis quantities x_1 (centre), x_0 (crossing), x_2 (recovered abscissa), together with the knob Q and the reflected “square” point $+P$, close into one loop (blue): fixing the curve x_1 and any two of $\{Q, x_0, x_2\}$ forces the rest. The yellow triangle $x_1 x_0 Q$ is one face of it. The degree-three curve provides the incidence but no extra freedom: the square stays two degrees of freedom (the two spokes), the curve buying nothing.

42.4 Why seven and not nine

Theorem 23 (Existence is a rank question, not a freedom count). A 3×3 magic square of nine squares at centre k^2 exists iff, for some two square-points $P, Q \in E$, both $P \oplus Q$ and $P \ominus Q$ are again square-points (integer legs on the same circle). The two knobs place P, Q freely; the group law then determines $P \oplus Q, P \ominus Q$ with no freedom left, so the question is whether E carries square-points at those two forced positions — a property of the rational points of E (its Mordell–Weil rank), not of the number of parameters.

This is the precise sense in which the two-parameter count of Section 41 does not settle the problem: after the group law consumes the freedom, existence passes to the arithmetic of E . For $k = 425$ the chosen P, Q make one of the two forced combinations a square-point (the both-square spoke 97104, cells $289^2, 527^2$) while the other is not ($\gamma_3 = 180096$ gives 23^2 and the non-square 360721): four of the six dependent cells square, the seven-square record of Theorem 20. A nine-square magic square is a single centre whose curve carries square-points at *both* forced positions at once; none has been found, and by the boundary obstruction (Theorem 18, the forbidden apex $s = d$ at the irrational radius $k = s\sqrt{2}$) the forced points are pushed toward a place the lattice does not reach.

remark 41 (Rank enlarges the supply, not the freedom). Rank and degrees of freedom are different quantities, and it is worth stating that they do not trade against one another. The Mordell–Weil rank counts the independent rational points of E — the *supply* of square-point candidates; a higher rank means more points from which to draw P, Q and better odds that $P \oplus Q$ and $P \ominus Q$ fall on square-points. But the square is still placed by exactly two choices, P and Q , after which the group law fixes $P \oplus Q, P \ominus Q$ with nothing left to adjust (Section 42.3). Hence even on a curve of large rank the problem remains *two* degrees of freedom: rank changes how *likely* the two forced points are to be square, never how *many* free points there are. The wall is not a shortage of freedom but the simultaneous squareness of two forced combinations of two free ones — and no amount of rank turns the two knobs into three. This is why the two-parameter count of Section 41 and the arithmetic (rank) of Theorem 23 are consistent, not competing: the geometry fixes the freedom at two, the arithmetic decides whether those two can be met, and passing to the elliptic curve — of whatever rank — buys extra candidates but never an extra knob.

Moreover the two are coupled by the circular relation of Section 42.3, and this is what *caps* the value of the extra candidates. A high rank supplies more rational points, but they help only if the forced images $P \oplus Q, P \ominus Q$ — already determined by the two free points — happen to fall on them, *closing* the loop. One cannot place the forced points on the new candidates independently; they must close at the end. So a large rank raises the density of possible landing sites without loosening the closure that binds them to the two choices: more candidates, each still obliged to close the same two-freedom loop. The geometry (closure) and the arithmetic (supply) do not add — they multiply against each other, and the circular relation is their meeting point: every candidate the rank offers must still return through the loop, so the abundance is spent on *odds*, never on *room*.

And the odds themselves lie *inside* the two degrees of freedom, never beyond them: every candidate the rank supplies is reached by the same two free choices, so meeting the six squareness conditions of Theorem 22 stays six constraints on a two-dimensional family — overdetermined by four — at every rank. Rank changes only how densely that two-dimensional family is populated by rational points, never its dimension. Six conditions to be fitted into two degrees of freedom is therefore the invariant wall: the whole problem is to place six square-demands inside a two-parameter family, and no arithmetic abundance, of any rank, enlarges the two or lowers the six.

remark 42 (The six conditions are coupled, not independent). “Six conditions in two degrees of freedom” must not be read as six independent events. The two free choices are A, D (taken square); the six conditions then fall on the *dependent* cells, each an integer-linear form in (k^2, A, D) ,

$$I = 2k^2 - A, \quad F = 2k^2 - D, \quad C = A + D - k^2, \quad G = 3k^2 - A - D, \quad B = 4k^2 - 2A - D, \quad H = 2A + D - 2k^2$$

(Theorem 22). These six have Jacobian rank *two* in (A, D) : only I depends on A alone and only F on D alone, while C, G, B, H depend on both. Hence no three of them are independent of the other three — any triple that involves both knobs already fixes (A, D) and therefore the rest, and the relations $I + F + C = 3k^2, C + G = 2k^2, B + H = 2k^2$ cross every partition. None of the six can

be tuned on its own — all are locked the instant A, D are chosen — so they are not six separate chances but one rigid system driven by two knobs. Hence they are strongly *coupled*: moving A, D to make one dependent cell a square shifts the other five as well, and generally pushes some off a square. There is no product of six probabilities to raise independently; there are two choices that fix all six values together, which is exactly why success is rarer than a naïve independence estimate would suggest — and why the record sticks at four of the six (Theorem 20) rather than climbing cell by cell. The circular relation of Section 42.3 is this coupling: the last cells fight the first rather than add to them, and each square gained costs freedom the remaining ones needed. The six sit inside a single closed convex, the two triples interlocked rather than independent (Figure 11); the algebraic form of the same coupling is Figure 12.

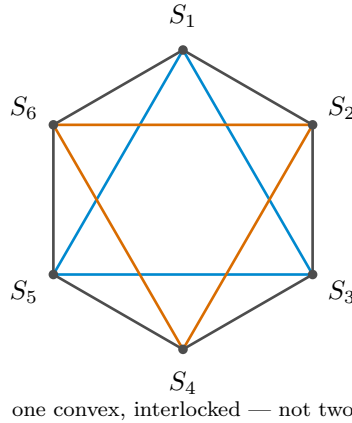


Figure 11: All six lie inside *one* closed convex. The six conditions are the vertices of a single hexagon; the two inscribed triples $\{S_1, S_3, S_5\}$ (blue) and $\{S_2, S_4, S_6\}$ (orange) *interlock* and cannot be pulled apart into two independent convex pieces. This is the coupling made visible: rank two in (A, D) (Remark 42) means (S_1, S_2, S_3) always depends on (S_4, S_5, S_6) , so the six squareness demands live in a single closure, never split three-and-three into separate ones. The algebraic form of the same fact — all six functions of the two knobs A, D — is Figure 12.

42.5 A network formulation and a bounded search

Reading the eight border cells as the fully coupled network of Figure 12 turns the problem into a two-input relaxation. Fix the centre k and inject one square $A = a^2$; the single free knob D then *settles* every other cell through the row/column/diagonal sums,

$$I = 2k^2 - A, \quad F = 2k^2 - D, \quad C = A + D - k^2, \quad G = 3k^2 - A - D, \quad B = 4k^2 - 2A - D, \quad H = 2A + D - 2k^2,$$

and one never needs the raw values — only the normalised weights $w_i = \text{cell}_i/k^2$, which satisfy $\sum_i w_i = 8$ (Proposition 48). The scoring function counts how many weights are rational squares (perfect-square cells), with the nine entries required *distinct* — the degenerate solution (three copies each of $205^2, 425^2, 565^2$) is excluded.

```
from math import isqrt
def score(k, A, D):
    e = k*k
    cells = [A, 4*e-2*A-D, A+D-e, D, 2*e-D, 3*e-A-D, 2*A+D-2*e, 2*e-A] #
            8 border cells
```

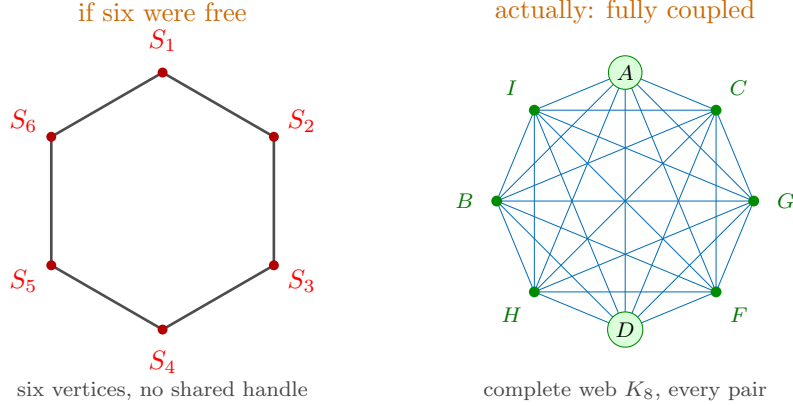


Figure 12: The six on one closed hexagon, and the full dependency web. *Left*: were the six conditions free they would be independent vertices with no shared handle. *Right*: as a coupled system the eight border cells form the *complete* graph K_8 : a generic change of the two knobs (A, D) moves *all eight* cells at once, so every pair co-varies and every node has degree seven. (Writing $F = 2k^2 - D$, $I = 2k^2 - A$ in the (A, D) basis makes F, I look tied to a single knob, but the row and column sums re-adjust every cell, so F and I move with the rest — the graph is fully connected, not near-connected.) One fixes (A, D) and the octagon settles the other cells through its relations; normalised by k^2 the eight vertex-weights sum to 8 (Proposition 48). This is why no three cells can be freed from the other three (Remark 42), and the record settles at four (Theorem 20) rather than climbing cell by cell.

```

if any(v <= 0 for v in cells):    return -1
if len(set(cells + [e])) != 9:    return -1          # distinct
    entries required
return sum(isqrt(v)**2 == v for v in cells)          # how many
    are squares

```

Sweeping the free knob $D = d^2$ for the input $A = 205^2$ at $k = 425$ settles at $D = 527^2$ with score 6 of 8 (seven of nine), recovering the record. A bounded scan over all centres and square inputs finds nothing better: for $k \leq 150$ the maximum is 5 of 8 (six of nine), and 425 is the smallest centre reaching seven of nine; no centre examined, here or elsewhere, exceeds it. As throughout, this is empirical — “none better below the bound” — and not a proof: an eighth or ninth square would require the coupled network to settle two more forced cells onto squares at once, the simultaneous landing the two-input coupling makes so rare (Remark 42).

The naive sweep is $O(n^3)$ in the bound (centres times two inputs). The objective is discrete — squareness is not differentiable and the solutions are isolated integer points, so gradient or backpropagation methods do not apply — but the arithmetic gives a clean $O(n^2)$ replacement: the companions I, F are square only when A, D lie in a both-square pair $2k^2 = x^2 + y^2$, and those members are found in one $O(k)$ pass per centre and then combined. This reaches $k \leq 5000$ in a few seconds, where the naive scan struggles past $k = 250$. The ceiling holds throughout: the best is always seven of nine, and the centres attaining it are exactly the multiples $425m$ — scale copies of 425 (Proposition 49), no primitive centre beyond 425 appearing and nothing above seven of nine found. Still empirical, still not a proof.

remark 43 (The two knobs are a corner and an edge). The two free cells are not interchangeable. In the settlement formulas A carries coefficient 2 ($B = 4k^2 - 2A - D$, $H = 2A + D - 2k^2$) while D

carries 1, so A occupies a *corner* of the grid and D an *edge*. Consequently, entering the network (Section 42.5) at a given square value recovers a square’s full score only if that value can sit at the A -corner — i.e. only if it is a corner cell of the target square. For the record at $k = 425$ the three square-valued corners each root a $7/9$ settlement, while the three square-valued edges fall short:

root	205^2	373^2	565^2	527^2	289^2	23^2
position	corner	corner	corner	edge	edge	edge
best	$7/9$	$7/9$	$7/9$	$6/9$	$6/9$	$4/9$

So the swap $A \leftrightarrow D$ is not a symmetry: corner and edge are distinct roles, and rooting at an edge value drops the score. Crucially, re-rooting does not change the *algebra* — A is always the coefficient-2 corner and D the coefficient-1 edge — it only changes which *value* is placed at the fixed corner. The root’s influence is therefore *geometric* (a value-to-position assignment), not algebraic: a value that is a corner cell of the target square fits the corner and keeps the score, while an edge value forced onto the corner yields a different, lower settlement. A square looks robust to re-rooting only when, like the record, it carries several square-valued corners.

Rooting at the corner is exactly the hypotenuse view of Section 42.6. The corner A , the centre E , and the opposite corner I form the diagonal

$$A = (d - s)^2, \quad E = k^2 = s^2 + d^2, \quad I = 2k^2 - A = (d + s)^2,$$

so the corner–centre–corner triple *is* the Pythagorean triple (s, d, k) , and choosing the corner value chooses its leg pair $s = \frac{\sqrt{I}-\sqrt{A}}{2}$, $d = \frac{\sqrt{I}+\sqrt{A}}{2}$ — the $\frac{q-p}{8} \cdot 4$, $\frac{p+q}{8} \cdot 4$ recipe. For 425, corners 205^2 , 565^2 give $s = 180$, $d = 385$ with $180^2 + 385^2 = 425^2$. Only a value that is $(d - s)^2$ for some leg pair of k — a genuine hypotenuse cell — can hold the corner and keep the diagonal all-square; an edge value belongs to the middle-row spoke (a different triple) and breaks the diagonal when forced to the corner.

One might still hope to see the swap on the curve as “which point is doubled,” since the corner A carries coefficient 2. But that coefficient is *convention-dependent*, not an intrinsic doubling. The four gaps of the 425 square, $\{41496, 97104, 138600, 180096\}$, are written $\{|p|, |q|, |p+q|, |2p+q|\}$ with a coefficient 2 in the corner–edge labelling $(p, q) = (-138600, 97104)$, and equally $\{g_1, g_2, g_1+g_2, g_1-g_2\}$ — with *no* doubling — in the symmetric two-diagonal labelling $(g_1, g_2) = (138600, 41496)$. The same four gaps, one labelling with a 2 and one without. So there is no intrinsic doubled point: the corner/edge asymmetry is a labelling of grid positions, not a group-law doubling on the curve, which is why the honest picture is the two-layer Figure 13 and not a single curve diagram.

remark 44 (The network closes the symmetry; squareness rides on top). The central symmetry — opposite cells summing to $2k^2$, every line to $3k^2$ — is not something to be arranged but something the K_8 network *enforces*: for *any* value placed at A the coupling settles $I = 2k^2 - A$ and then, through D , all remaining cells, always preserving the antipodal pairing and the magic sums (Section 42.3). Changing A therefore never breaks the symmetry; the network re-closes it across all eight cells. What A changes is the *arithmetic* layered on top — how many cells are perfect squares (the score) and whether A holds a genuine corner/hypotenuse cell (Remark 43). So the 3×3 structure separates cleanly into two levels: a central symmetry the network always closes, and a squareness condition the two free choices can only partly meet — the former guaranteed, the latter the open difficulty.

42.6 What selects a centre: mod 8, closure, and the hypotenuse

Three things are commonly blamed for why 425 works — the number of square roots it offers, its size, and its factor count. None is the operative reason; what a centre must provide is structural.

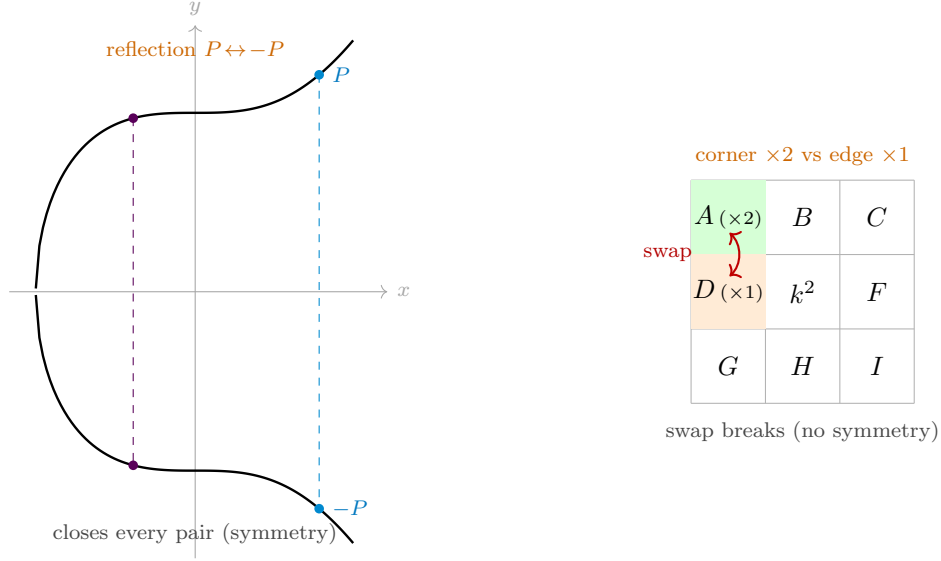


Figure 13: Two layers. *Left (on the curve)*: the elliptic curve’s reflection across the x -axis, $P \leftrightarrow -P$, closes every antipodal pair (cell + companion = $2k^2$) — a genuine symmetry that always holds, treating the diagonal pair and the middle-row pair identically. *Right (in the grid)*: the two free cells sit at a corner (A , coefficient 2) and an edge (D , coefficient 1); swapping them is *not* a symmetry, so an edge value forced to the corner breaks the score. The closing symmetry lives on the curve; the breaking asymmetry lives on the grid — different layers, which is why the swap cannot be shown “breaking” on the curve itself.

Not the number of roots. A centre may present many square inputs that each settle to a decent score, yet they do not combine: each is one two-degree-of-freedom settlement (Section 42.3), and the favourable cells of different settlements live in different closures. For $k = 442$ five input roots reach six of nine across two distinct squares, and none can be merged to exceed it. Abundance of roots is not progress.

Requirement 1: primitive mod 8. By Theorem 21 and Corollary 18 a primitive square needs k odd, so every cell is $\equiv 1 \pmod{8}$; an even k merely rescales a smaller one. The quick signature is $k^2/8$: an odd k gives fractional part $\frac{1}{8}$ (e.g. $425^2/8 = 22578.125$), an even k gives 0 or $\frac{1}{2}$ (e.g. $422^2/8 = 22260.5$, since $422^2 \equiv 4$). Only the odd centres are primitive.

Requirement 2: the roots must close. Beyond mod 8, the chosen roots must assemble into the closed two-parameter configuration of Section 42.3 with the forced cells also square. This closure — not the supply of roots — is decisive, and for the full nine it is the open problem.

Choosing a centre: the hypotenuse rule. The raw material for closure is the set of *both-square spokes*, and each is a Pythagorean triple. A pair of cell-roots (p, q) around k has legs

$$s = \frac{q-p}{2} \text{ (half-difference),} \quad d = \frac{p+q}{2} \text{ (mean),} \quad s^2 + d^2 = k^2,$$

so (s, d, k) is a right triangle and $p = d - s$, $q = d + s$. A both-square spoke exists at k iff k is the *hypotenuse* of an integer right triangle; a centre supplies *many* spokes iff it is a hypotenuse in many ways — which happens exactly when k carries primes $\equiv 1 \pmod{4}$.

Why $425 = 5^2 \cdot 17$. Both 5 and 17 are $\equiv 1 \pmod{4}$, so 425 is a hypotenuse seven ways; the seven triples and their both-square pairs are

s	d	pair ($d-s, d+s$)	s	d	pair
297	304	(7, 601)	180	385	(205, 565)
255	340	(85, 595)	119	408	(289, 527)
200	375	(175, 575)	87	416	(329, 503)
65	420	(355, 485)			

The pairs (205, 565) and (289, 527) are the two both-square spokes of the record square.

Worked example (reading the legs off a pair). Take $(p, q) = (17, 25)$. Dividing by 8 keeps the mean over the eight border vertices, and multiplying by 4 restores the four-above/four-below split, so the legs come out as

$$\frac{q-p}{8} = \frac{25-17}{8} = 1, \quad 1 \cdot 4 = s = 4; \quad \frac{p+q}{8} = \frac{17+25}{8} = 5.25, \quad 5.25 \cdot 4 = d = 21.$$

Then $s^2 + d^2 = 4^2 + 21^2 = 457$, which is *not* a perfect square: 457 is prime and $\equiv 1 \pmod{4}$ but is not itself a hypotenuse, so (17, 25) seeds no spoke. A second pair (35, 43) behaves the same way: $\frac{43-35}{8} \cdot 4 = 4 = s$ and $\frac{35+43}{8} \cdot 4 = 39 = d$, whence $s^2 + d^2 = 4^2 + 39^2 = 1537$, again not a perfect square — and here $43 \equiv 3 \pmod{8}$ (equivalently $\equiv 3 \pmod{4}$), so the prime 43 is itself no hypotenuse. (The same recipe on (205, 565) gives $\frac{360}{8} \cdot 4 = 180 = s$ and $\frac{770}{8} \cdot 4 = 385 = d$ with $180^2 + 385^2 = 425^2$ — a valid spoke.)

Failing case (a prime not $\equiv 1 \pmod{4}$). A centre built only from primes $\equiv 3 \pmod{4}$ fails outright. Take $k = 21 = 3 \cdot 7$: it is odd, so it *passes* the mod-8 test ($21^2/8 = 55.125$, cells $\equiv 1 \pmod{8}$), yet $21^2 = 441$ is a sum of two squares only trivially, $441 = 21^2 + 0^2$. No leg pair $s, d > 0$ solves $s^2 + d^2 = 21^2$, so 21 is the hypotenuse of nothing and seeds no both-square spoke — the search never even starts. This is the decisive difference from 425: passing mod 8 (odd) is necessary but useless without a prime $\equiv 1 \pmod{4}$ to make k a hypotenuse. The intermediate case $851 = 23 \cdot 37$ carries one such prime ($37 \equiv 1 \pmod{4}$) but only one, so it is a hypotenuse a single way — one spoke, too few to close.

How many spokes: count the primes $\equiv 1 \pmod{4}$ (not $\pmod{8}$). The number of both-square spokes grows with the distinct primes $\equiv 1 \pmod{4}$ dividing k (each raises it, higher powers raise it more). Crucially it is $\equiv 1 \pmod{4}$, not $\equiv 1 \pmod{8}$:

k	factors	primes $\equiv 1 \pmod{4}$	primes $\equiv 1 \pmod{8}$	spokes	best
5	5	5	—	1	0/9
25	5^2	5	—	2	6/9
65	$5 \cdot 13$	5, 13	—	4	5/9
425	$5^2 \cdot 17$	5, 17	17	7	7/9
1105	$5 \cdot 13 \cdot 17$	5, 13, 17	17	13	6/9
5525	$5^2 \cdot 13 \cdot 17$	5, 13, 17	17	22	7/9
2009	$7^2 \cdot 41$	41	41	1	0/9

The spoke count follows the $\equiv 1 \pmod{4}$ column, never the $\equiv 1 \pmod{8}$ one: 65 has four spokes with *no* prime $\equiv 1 \pmod{8}$ (both 5, 13 $\equiv 5 \pmod{8}$), while 2009 carries the prime $41 \equiv 1 \pmod{8}$ yet has only one spoke. So the productive prime 5 — behind the 3-4-5 triangle — is $\equiv 5 \pmod{8}$, and a mod-8 filter would wrongly discard it. More distinct primes $\equiv 1 \pmod{4}$ give more spokes and a better chance to close; but more spokes is not more score (1105 has thirteen and reaches only 6/9) — the raw material grows, the closing does not follow.

Count is material; closure needs a compatible pair. A magic square uses not one spoke but *two*, placed on two lines through the centre: one on the corner diagonal $A-E-I$ and one on the middle row $D-E-F$, both all-square (Remark 43). These two are not free to be any of the spokes: their gaps must satisfy the (p, q) relations of Theorem 19, so that the induced anti-diagonal and middle-column cells land where they must. Of the seven spokes of 425, only the pair $\gamma = 138600$ (diagonal) and $\gamma = 97104$ (middle row) closes; the rest do not fit together. So the spoke count is only how many candidate diagonals one *has*; the square needs a *compatible corner-diagonal-middle-row pair* to *close*, and the count does not guarantee one — 1105 (thirteen spokes) and 32045 (forty) offer more material yet no closing pair beyond 6/9, while 425 (seven) does.

A narrowing heuristic (empirical; one data point). One may ask not for many spokes but directly for the record. The tight rule $k = n^2 P$ with $P = n^2 - 8$ prime and $P \equiv 1 \pmod{8}$ (a square times a single prime) is far more *selective* than the spoke count: up to 3000 it flags only $\{425, 2009\}$, of which 425 is the 7/9 centre — precision $\frac{1}{2}$, against the spoke rule which floods with 6/9 centres. But it still fires on 0/9 duds (2009, and 5913, 13673, ... further out), so its precision decays; and it misses the scale copies 850, 1275, ... that also reach 7/9 (recall $\frac{1}{7}$). Above all, every known $\geq 7/9$ centre is a multiple $425m$ of a *single* primitive square, so with one positive example no rule can be separated from a coincidence: $25 - 8 = 17$, $k = 25 \cdot 17$ fits 425 exactly, but so would many descriptions of one point, and the rule also fires on 2009. We record it only as a narrowing filter around the known case, not a validated predictor; the operative, theorem-backed quantity remains the count of primes $\equiv 1 \pmod{4}$ (spokes), with closure into the record still unforced.

The boundary. These are *necessary* conditions — odd $k \pmod{8}$ and k a rich hypotenuse (enough both-square spokes) — not sufficient ones. Many hypotenuse spokes still need not *close* into a nine-square configuration, and whether they ever do is unresolved. What is settled is the negative direction: a centre lacking primes $\equiv 1 \pmod{4}$ has too few spokes to close at all, so neither size nor factor count helps. It is the hypotenuse structure, not the magnitude of k , that matters.

Acknowledgements

The implementation was developed and tested interactively; the verification figures above are from exhaustive computation over the stated range.